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DIGITALISING NATURE

Sociotechnical Dynamics in the Making of Digital Representations

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To my mother

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Abstract

Digital technologies are increasingly being positioned as central to addressing grand societal challenges such as climate change and biodiversity loss, particularly in relation to the management and protection of natural environments. Previous Information Systems (IS) research has however largely focused on artificial objects and organisational settings, leaving the phenomenon of digitalising natural environments underexplored. Understanding how sociotechnical interactions shape this process is therefore both opportune and necessary. In this dissertation, I develop a sensitising device comprising the object of digitalisation, data, and digital representations to examine and understand this distinct phenomenon. Drawing on both qualitative and design science research situated in Swedish forestry, this study investigates how this digitalisation process unfolds.

In doing so, thesis shows that the digitalisation of natural environments is a deeply sociotechnical process shaped by the distinct characteristics of natural objects and the competing interests of multiple stakeholders, which actively influence what can be captured as data and digitally represented. Where multiple perspectives on data coexist within the same digitalisation effort, the perspective that dominates shapes the trajectory and outcomes of the process. Achieving a more comprehensive digitalisation of natural environments therefore requires moving from purely commodity-oriented views of data towards more relational perspectives.

Taken together, this thesis makes novel contributions to IS research and practice. Theoretically, it extends scholarly attention beyond artificial objects to natural environments as a distinct object of study, and enriches existing concepts of digitalisation, digital transformation, and digital innovation through a novel empirical context. It also develops and applies a sensitising device that provides an analytical lens for studying how this process unfolds in practice. For practitioners, the thesis offers implications for how managers can adopt pluralistic perspectives on data in digitalisation efforts, how policy and public investment can support more inclusive digital systems, and how diverse actors can be brought together to foster participatory planning of natural environments.

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Preface

This dissertation consists of a cover manuscript and four research papers. The purpose of the cover manuscript is to explore the digitalisation of natural environments and the complexities it entails. It offers a conceptual device that enables analysing how the digitalisation of natural environments is enacted in practice at the organisational, industry, and societal levels. The four research papers, presented after the cover chapter, address distinct yet interrelated aspects of this digitalisation. The research papers included in this thesis are:

Paper 1: Torto, M. & Kristofersson, A. (2023). Digitally Transforming in a Traditional Industry: A Senior Management Perspective. *Proceedings of the 56th Hawaii International Conference on System Sciences (HICSS).*

Paper 2 : Torto, M., Nylén, D., & Sandberg, J. (2025) Digital Innovation and the Representation of Natural Objects: The Role of Physicality. *Proceedings of the Thirty-first Americas Conference on Information Systems*

Paper 3: Torto, M., Nylén, D., Sandberg, J., Nordström, M., Cartling Wallén, M., & Södergren, F. Establishing Data Spaces for Innovation: Investigating Challenges and Responses in the Forestry Industry. (In manuscript)

Paper 4: Torto, M., Lindberg, A., Nylén, D., & Sandberg, J. Enabling Environmental Justice through Digital Twins: A Design Theory. (In manuscript)

1. Introduction

Digital technologies have become a prominent and commonplace feature of contemporary life transforming how we live, how businesses operate, and how entire sectors function (Yoo et al., 2024; Wibowo et al., 2022; Lyytinen, 2022; Legner et al., 2017). Beyond reshaping value chains and industries, digital technologies are increasingly seen as instrumental in addressing complex societal challenges (Faik et al., 2020; Selander & Jarvenpaa, 2016). In the health sector, for instance, artificial intelligence (AI)-enabled technologies are significantly improving disease diagnoses and extending life expectancies (Lebovitz et al., 2021), whereas in the education sector virtual and distance learning technologies are opening up opportunities for students who would formerly have been excluded (UN, 2020). More recently, this transformative potential has been directed towards the domain of natural environments, where the stakes are particularly high.

It is widely recognised that environmental crises such as climate change and biodiversity loss are at the centre of global political and economic discourse (IPCC, 2023). For several decades, growing concerns about the deterioration of natural environments have highlighted how central they are to sustaining both ecological and economic life (IPCC, 2023; Andersen & Wicken, 2021). Natural environments are simultaneously vulnerable to the effects of climate change and central to addressing it. For example, forests act as global carbon sinks removing carbon dioxide from the atmosphere (Luyssaert et al., 2008), and along with other natural environments play important roles in improving human well-being and in biomass production (Skogsstyrelsen, 2022).

Given the continued rise in greenhouse gas emissions, concerns about these environmental crises are signs of a grand societal challenge with profound and wide-ranging consequences on human livelihoods and communities (IPCC, 2023). Digital technologies have been proposed as a key part of the response to address these concerns. By modelling and monitoring environmental conditions such as climate patterns, and biodiversity, and pollution, digital technologies offer new possibilities to understand and forecast environmental change (Mondejar et al., 2021; OECD, 2024). As a result, digitalisation has become a fixture in the agendas of national and international forums as a pathway towards more inclusive and sustainable management of natural environments

(Barteková & Börkey, 2022). The Information Systems (IS) discipline is well-positioned to contribute to these efforts.

IS research has long examined how digital technologies reshape the contexts in which they are used, and information systems have been widely acknowledged as key resources in realising sustainable transformations (Ixmeier et al., 2024) and addressing sustainability-related challenges (Seidel et al., 2013; Kotlarsky et al., 2023). Scholars have for instance explored how organisations use digital technologies such as satellite imagery and remote sensing scanning to monitor forest conditions (e.g. Schoormann et al., 2023). Similarly, digital systems have been used to collect better quality data to capture and assess the state of natural environments (Gholami et al., 2016). In this regard, digital technologies are seen as key enablers of social and economic sustainability, extending the benefits of digitalisation to different communities and contexts (Bork et al., 2024)

IS research has begun to examine the digitalisation of natural environments, although attention to this phenomenon remains limited. Some exceptions still exist. For instance, Monteiro and Parmiggiani (2019) develop the concept of 'synthetic knowing' to describe how digital representations of marine phenomena such as fish and biomass come to constitute knowledge in Internet of Things (IoT)-focused contexts. Similarly, Østerlie et al. (2012) examine how natural phenomena such as seabeds shape what can be captured as data and digitally represented, showing how sensors play an integral role in creating the materiality of physical phenomena. Finally, Østerlie and Monteiro (2020), explore how physical situated work practices are replaced with remote work practices using digital representations in the case of sand monitoring. Still, scant attention has been paid to how the distinct characteristics of natural objects shape the digitalisation process. In this thesis, I set out to generate an increased understanding of the digitalisation of natural environments as a distinct phenomenon that unfolds across individual, organisational, and societal levels, and to illustrate this phenomenon.

1.1 Problem Statement and Research Objectives

The digitalisation of natural environments presents distinct theoretical and practical challenges that distinguish it from the digitalisation of phenomena predominantly studied in IS research. Historically, the IS field has focused its enquiry on organisational processes or artificial objects which are human-made and designed with specific purposes in

mind. In contrast, natural objects behave in accordance with biological and ecological rules and possess intrinsic agency and autonomy (Moreno & Etxeberria, 2005), independent of human intervention (Misselhorn, 2015). As natural objects are active entities (Sultan et al., 2022) the process of digitalising them cannot be understood through the same conceptual lens applied to organisational processes or human-made objects. While scholars have explored how digital technologies can be used in natural environments (Alcocer et al., 2022), from remote sensors to citizen science platforms (Blumstein et al., 2011; Lotfian et al., 2026), these environments have largely been treated as passive settings for the technologies, rather than as focal objects whose specific characteristics shape the digitalisation process itself. As a result, important questions about what the digitalisation of natural environments actually entails remain unanswered.

Addressing these critical questions requires recognising that digitalising natural environments involves decisions that are as deeply social and political as they are technical. Decisions regarding what aspects of a natural environment are captured as data, how that data are governed, and how natural phenomena are eventually represented are all shaped by the interests, values, and goals of the stakeholders involved. These decisions do not occur in isolation but rather in sociotechnical systems where actors interact to generate emergent outcomes. Understanding the interactions between actors, digital technologies, data, and natural objects during the digitalisation process is therefore important. However, these interactions remain systematically under-examined in IS research. Against this backdrop, this thesis addresses the following research question:

How do sociotechnical interactions shape the digitalisation of natural environments?

Answering this research question requires a conceptual framework that accounts for the distinct nature of the phenomenon. This thesis therefore develops a sensitising device to analytically examine the actual objects of digitalisation, the role of data, and digital representations in the digitalisation of natural environments (e.g., Jones, 2019; Mikalsen & Monteiro, 2021; Alaimo & Kallinikos, 2024). In investigating the interplay between these three elements, this thesis sets out to study how what counts as an object of digitalisation shapes which data gets collected and for whom, which in turn influences how natural environments are digitally represented. In doing so, the thesis uncovers how the

digitalisation of natural environments unfolds in practice and what it means for the different stakeholders involved.

1.2 Research Approach and Contributions

To empirically investigate this research question, the thesis is situated within the empirical setting of Swedish forestry. Swedish forestry is a large and historically significant industry that has been central to the national economy for over a century (Skogsindustrierna, 2025; Fridman & Danell, 2024). Beyond its economic contributions through exports, Swedish forestry serves multiple purposes and is important to a wide range of actors and stakeholders (Lindahl & Westholm, 2012). It is a setting rich in cultural, historical, and social values, which means that there are many competing and often incompatible interests and perspectives on how forests should be used and managed (Sandström et al., 2016). This thesis therefore collates four papers that examine the digitalisation of natural environments within this context. In these papers I have employed methods focused on qualitative case studies and a design science approach.

Specifically, Paper 1 presents a qualitative case study at the industry level and investigates the factors that can contribute to and inhibit the success of digital transformation efforts. Paper 2 explores the digitalisation of natural environments at the organisational level, illustrating how a single firm supports forest management activities for customers by digitally representing aspects of the forest in a mobile application. This paper highlights the distinct characteristics of natural objects in shaping both the motivation for creating digital representations and what can ultimately be represented. Paper 3 examines cross-boundary data sharing across multiple data-driven research and development projects aimed at digitalising the forestry value chain and promoting industry-wide digital transformation. The paper identifies the capabilities needed to engage in a data space and innovate within them and explores how actors balance tensions between openness and control that arise in data-driven innovation. In doing so, it reveals the sociotechnical interactions that occur when industry actors share data as part of broader digitalisation efforts. Paper 4 raises critical questions about how digital systems for decision-making around natural environments can be exclusionary and are designed to prioritise the interests of dominant stakeholders while sidelining others. Drawing on design science research, the paper offers normative prescriptions on how systems, such as digital twins, can be designed to enable participatory planning of natural environments.

Taken together, this thesis provides novel insights into the value of advancing IS research to engage with the digitalisation of natural environments as distinct phenomenon. In doing so, it not only extends the empirical scope of IS research but also enriches its theoretical concepts and contributes to the cross-disciplinary conversation that this phenomenon demands. For practitioners, the thesis highlights the importance of adopting pluralistic perspectives on data to enable a more inclusive digitalisation process that brings diverse actors together.

1.3 Thesis Structure

This thesis consists of four research papers and a cover manuscript, which is structured as follows:

Chapter 1 introduces the thesis and situates the study within broader debates about digitalisation of natural environments. It presents the research problem and describes the need for IS research to engage with this phenomenon, before outlining the research question and objectives, and the approach taken to address them.

Chapter 2 examines the digitalisation of natural environments as the overarching phenomenon studied in this thesis. It discusses the importance of natural environments, and the growing role that digital technologies play in how they are monitored, managed, and engaged with. The chapter also reviews relevant IS scholarship on digitalisation and the related concepts of digital transformation and innovation.

Chapter 3 develops the conceptual framework used in this thesis. It introduces objects of digitalisation, data, and digital representations as three key elements that form a sensitising device for studying the digitalisation of natural environments.

Chapter 4 describes the research design and methodology, detailing the philosophical and interpretive foundations of this thesis. It explains the qualitative case study and design science research approaches used across the appended papers, as well as describing the data collection and analysis processes. This chapter also provides an overview of the Swedish forestry industry as the empirical setting of this thesis, and highlights researcher positionality and ethical considerations.

Chapter 5 summarises the four appended papers in this thesis. It outlines the research question, area of concern, theoretical framing, and key findings of each paper.

Chapter 6 discusses the main findings of the thesis and elaborates on what they collectively illuminate about the digitalisation of natural environments.

Chapter 7 concludes this thesis by summarising the theoretical contributions to IS research and implications for practice, reflecting the limitations of the dissertation, and identifying directions for future research.

2. Digitalisation of Natural Environments

Digital technologies have become more pervasive than ever. Emerging, newfangled technologies such as big data analytics, the Internet of Things (IoT), and artificial intelligence (AI) have been widely framed as central to addressing a range of societal and organisational challenges (Plank et al., 2026). Rather than operating as pure standalone tools, digital technologies are now deeply embedded in daily routines and broader social structures, actively shaping how work is organised, and realities are created (Baskerville et al., 2020). Information Systems (IS) research has long been concerned with understanding the role of digital technologies in such contexts. Historically, the field has focused on IT artifacts[1] – applying IT to enable or support tasks embedded within a context – and how they are constructed, used, and evolve over time as well on the organisational and societal impacts that emerge as they are embedded into different contexts (Benbasat & Zmud, 2003, p.186). More recently, attention has shifted towards digital artifacts as central organisational resources, examining how they are positioned and bundled in organisations' ever-changing business models to generate value and competitive advantage (Kallinikos et al., 2013; Faulkner & Runde, 2019). Despite this broadening scope, the predominant focus has been on artificial objects and sociotechnical systems within organisational settings. Only in more recent years have some scholars begun to extend their attention and research to other focal objects, including natural environments (e.g. Østerlie et al., 2012; Monteiro & Parmiggiani, 2019; Østerlie & Monteiro, 2020). This shift has been driven in part by the growing notability of grand societal challenges such as climate change and loss of biodiversity, which demand new forms of research insights, knowledge, and innovations (Bruynseels, 2020).

At a global level, the importance of digital technologies and data in responding to these challenges have been formally recognised. For example, the United Nations (UN) has highlighted how digital solutions help build capabilities for stewarding ecological systems as well as the role that data-driven approaches play in planning and informing decisions about biodiversity loss and ecosystem management and conservation

[1] Orlikowski & Iacono (2001) in their seminal work on establishing a core identity for the IS discipline highlight how the IT artifact should be at the forefront of IS research.

(UNDP, 2025). These developments signal a potential for IS research to extend beyond organisational settings and explore how digital technologies can be used to gain insights into natural environments and address associated challenges.

The following sections describe the importance of natural environments and expand on how digitalisation can offer substantial opportunities and improve sustainability practices.

2.1 Natural Environments and Digital Technologies

Natural environments are the bedrock of human well-being and are indispensable to societies. They consist of complex interconnected ecosystems that play a vital role in human survival and maintaining ecological balance (Abe, 2023; Edwards & Abivardi, 1998). That said, there are multiple worldviews and underlying assumptions on the importance of nature. Societies and individuals have expressed the value and importance of natural environments in many ways (Himes et al., 2024). On the one hand, natural environments are understood to have intrinsic value, where entities are valuable in themselves and for their own sake (Vilkka, 1997). Here, natural environments are seen as having agency and holding value independent of the interests or judgements of humans (Silvertown, 2015; Himes et al., 2024). Intrinsic value is often regarded as being particularly relevant for legitimising environmental protection efforts and developing conservation plans (Sandler, 2012).

On the other hand, other scholars adopt a more anthropocentric perspective, which centres and justifies human conduct only in relation to human desires and motives (Hovardas, 2013). Within this view, natural environments are assigned instrumental value. Natural environments are thus understood as a means to a desired end (Sandler, 2012), such as improving human well-being and satisfying human preferences (Reyers et al., 2012; Hovardas, 2013). Accordingly, they are framed in relation to ecosystem services or as assets (Adams, 2014) with their value recognised in terms of utility and commercial or industrial worth (Vilkka, 1997). However, scholars argue that instrumental values should not be limited to only economic considerations but should encompass broader values, ranging from providing food for people to cultural, spiritual, and aesthetic aspects (Reyers et al., 2012; Maguire & Justus, 2008). Even so, both philosophies on the value and importance of natural environments have been critiqued for their limitations, with Himes et al. (2024) arguing that

intrinsic value may overlook pragmatic environmental management concerns and instrumental value risks alienating stakeholders or reducing emphasis environmental protection, as it is mainly attributed to human or economic gain.

These perspectives on the importance of natural environments do not exist in a vacuum, however, and are situated within broader discourses on sustainability, which have become more prominent over time. For the past few decades, concerns surrounding climate change and broader environmental crises have intensified, as societies increasingly realise the global risks these events pose to human populations and natural ecosystems (IPCC, 2023). Natural environments have been at the centre of these debates, not only as sites vulnerable to and affected by climate change, but also as key components of the ecosystems that support climate regulation (Luyssaert et al., 2008). In particular, they consist of biomes and habitats that act as carbon sinks and are seen as fundamental for absorbing carbon dioxide (CO_2) from the atmosphere and mitigating climate change (Regulation (EU) 2024/1991). At the same time, rising global greenhouse gas emissions, largely driven by human activities, have contributed to irreversible damages to atmospheric conditions and a rampant loss of biodiversity (IPCC, 2023). These environmental changes extend beyond adversely affecting ecology.

Climate change has increasingly produced economic and societal consequences that affect human societies and their livelihoods (IPCC, 2023). Climate variability and change have thus acted as catalysts for a portfolio of initiatives. Globally, United Nations (UN) member states have developed policies at national and sub-national levels[2] aimed at addressing these crises and pursuing net zero or low-emissions development strategies (UNESCO, 2020). As part of these efforts, the United Nations Development Programme (UNDP) has highlighted how digital technologies and innovations are expected to play an important role in informing decision-making for environmental protection and biodiversity conservation (UNDP, 2025). In this context, digital technologies are positioned to enable more targeted approaches to environmental management, including optimising resource utilisation (Adebayo et al., 2025). Across diverse ecosystems and research domains, scholars have explored how digital technologies can be deployed in natural environments to support such aims. For example, remote sensing technologies are seen as critical to the conservation and management of freshwater fish species and habitats. These technological advances and

[2] see Paris Agreement/ United Nations Framework Convention on Climate Change (UNFCCC)

data sources help map and monitor environmental phenomena in ways that were previously difficult to achieve (Kuiper et al., 2023). Similarly, ecologists have explored how sound and acoustic technologies can be used to study animals remotely and non-invasively, allowing for monitoring any changes in species' habitats and ecosystem health that may be caused by climate change or human activity (Blumstein et al., 2011). These approaches are helpful in detecting disturbances in natural environments, understanding associated biodiversity changes, and predicting future scenarios (Alcocer et al., 2022).

Digital technologies are used not just by researchers or organisations to monitor natural environments but are also used by the public as a way to interact with nature. One prominent example is broadly referred to as citizen science – defined as the involvement of citizens or non-experts in collaboratively creating new knowledge through various types of data (Fuchs et al., 2025) – where members of the public are able to engage more directly with nature using digital tools. Such initiatives exist across different parts of the world. In Brazil, for instance, digital technologies have been introduced in natural resource management projects in the Amazon basin to support forest and wildlife preservation efforts aimed at combating biodiversity loss and climate change[3]. Bird watchers, for instance, volunteer to collect data on bird sightings, and these data are used by researchers to study bird populations, migration patterns, and assess the impact of climate change on bird behaviour[4].

In this way, digital technologies play a prominent role both in addressing sustainability challenges and in how natural environments are monitored and engaged with. Given the complexity of societal challenges related to human interference with nature and the growing role of digital technology in such processes, studying the digitalisation of natural environments requires a conceptual vocabulary to account for the digital dimensions of this process. Here, IS research offers grounding for analysing the digitalisation of natural environments as it unfolds in practice. In the next section, I examine relevant IS scholarship that can clarify what digitalisation conceptually entails and the various levels at which it operates.

[3] MapBiomas project https://brasil.mapbiomas.org/en/o-projeto/
[4] Bird Citizen Science https://www.gvi.co.uk/blog/smb-bird-citizen-science-contributing-to-conservation-efforts-one-sighting-at-a-time/

2.2 Levels of Digitalisation

Leveraging digital technologies in relation to environmental phenomena is far from straightforward. The digitalisation of natural environments is a complex phenomenon that involves many different stakeholders including government bodies, natural resource extraction and management industries, local communities, and indigenous groups (Grimble & Wellard, 1997). Any digitalisation process is likely to gradually trickle down into the individual and organisational levels as the use of natural environments cuts across social, economic, and political units (Grimble & Wellard, 1997). As more diverse actors shift towards digitalising natural environments, fundamental questions arise about what it means to 'digitalise' and what such processes entails. IS research has been at the helm of examining the role of digital technologies across various contexts and settings, showing that introducing digital technologies is more than just a technological solution (Kane et al., 2019). The IS discipline thus offers key concepts to analyse and unpack the digitalisation of natural environments at various levels.

In IS research, *digitalisation* is distinguished from and expands on the concept of *digitisation*. Digitisation refers to the technical process of converting analogue signals into bitstrings (Tilson et al., 2010), producing information that can expressed in multiple ways (Brennen & Kreiss, 2016). In the process of digitising, physical objects or real-world phenomena are represented or converted into digital artifacts (Gradillas & Thomas, 2025). Classic examples include transforming previous nondigital and physical products such as books and print into e-books and digital magazines by embedding digital capabilities (Yoo et al., 2010; Nylén et al., 2014). As Tilson et al. (2010) note, however, using such digital technologies also gives rise to new social relationships and interactions among involved actors. Digitalisation thus extends the technical to encompass sociotechnical structures, processes, and interactions that have shifted as a result of digital technologies (Brennen & Kreiss, 2016). Reflecting this emphasis, IS scholars have defined digitalisation in various but related ways. Yoo et al. (2010), for instance, focus on the notion of digital technology's mediating role and define digitalisation as 'the transformation of socio-technical structures that were previously mediated by non-digital artifacts or relationships into ones that are mediated by digitized artifacts and relationships' (p. 6). More recently, Gradillas and Thomas (2025) describe digitalisation as 'the transformation of the socioeconomic environment through processes of digital artifact adoption, application, and utilisation' (p. 113). Taken together, these perspectives highlight that digitalisation has wide-ranging

implications that shape and influence multiple domains of social life (Brennen & Kreiss, 2016). To facilitate a more detailed multilevel account, in this thesis, I adopt the definition of digitalisation by Legner et al. (2017): 'the manifold sociotechnical phenomena and processes of adopting and using these [digital] technologies in broader individual, organizational, and societal contexts' (p. 301).

At the societal level, digital technologies have been introduced to boost social inclusion, and welfare services (Faik et al., 2020). For example, digital technologies have allowed for more digital health interventions for marginalised and disadvantaged communities (Ganju et al., 2016). Andrade and Doolin (2016) illustrate how technologies for social inclusion can be helpful in assisting refugees to integrate into new environments and regaining control over their displaced lives. At the individual level, digital technologies have become embedded in everyday life and are continuously shaping habits and social interaction. Through platforms such as social media, individuals are able to exchange information, sustain social relationships, and engage in new modes of communication (Matt et al., 2015). Beyond interpersonal interactions, digital technologies have also expanded opportunities and empowered citizens to participate in democratic processes and functions with new ways (OECD, 2022).

Digital technologies mediate a range of daily activities. According to the World Bank (2024), individuals use digital tools not only for remote work and schooling, but also for managing finances, accessing health services, conducting business, and participating in digital marketplaces. In this way, digitalisation restructures how individuals access services and engage with organisations. It equally enables organisations to integrate individuals directly into their value creation processes, while individuals receive services more tailored to their preferences and needs (Matt et al., 2015). In relation to natural environments, digital technologies are used in everyday life by enthusiasts to improve their experiences with nature across a wide array of activities (Webber et al., 2023). For example, in skiing, apps[5] have been created to provide real-time updates to skiers about weather conditions and slope openings and even offer real-time audio coaching for skiers at different skill levels. Another example of such activities is fishing, where anglers are able to use sonar technology that provide real-time high-resolution images of fish and underwater structures to increase the chances of a catch (Cooke et al., 2025).

[5] See https://getcarv.com

At the organisational and industry levels, digitalisation draws attention to how organisations manage and adapt to the changes that emerge when digital technologies are introduced into various contexts. In connection to natural environments, organisations involved in natural resource management and extraction must navigate with the sociotechnical transformations that result from digitalisation (Tretter & Burns, 2023). To understand how organisations respond strategically to disruptions caused by digital technologies, IS research has introduced the concept of 'digital transformation'.

The early discourse on digital transformation (DT) was characterised by ambiguity, with scholars debating whether it was a buzzword or fad (Baiyere et al., 2017). It has since come to be understood as a more concrete business transformation process undertaken predominantly by organisations (Chanias et al., 2019). At the height of the DT trend, researchers and practitioners struggled to converge on a shared understanding and definition of what DT was (Hanelt et al., 2021). Several definitions have reflected the evolving nature of this discourse. Vial (2019) defined DT as a process that aims to improve customer experiences, establish value creation pathways, develop new business models, and increase operational efficiency through the use of digital technologies. This definition places emphasis on digital technologies as playing a central role in DT (Hovorka & Müller, 2025). However, subsequent research has argued for a shift where DT engages with the more non-technological components of the process (Wessel et al., 2025). Additionally, Carroll et al. (2023) defined DT as 'the profound and accelerated transformation of business and organisational activities, processes, competencies, and models to fully leverage the opportunities of the digital era' (p. 347). Despite these efforts, the concept of DT continues to be used loosely, and often imprecisely, without pinpointing its defining characteristics (Chen & King, 2022).

With a lack of coherence in the literature, different disciplines outside of IS adopted and repackaged the DT concept to align with their specific goals (Markus & Rowe, 2023). Part of the confusion stems from a lack of conceptual clarity on the difference between DT and the parallel concept of IT-enabled organisational transformation (ITOT). Wessel et al. (2021) provide a useful distinction by highlighting that ITOT uses digital technologies to reinforce existing organisational identities and value propositions. ITOT thus focuses on improving operational efficiency by implementing digital technologies for internal processes (Baiyere et al., 2023). DT, by contrast, leverages digital technologies to redefine value propositions and offerings, and imbue or reshape organisational identity

(Wessel et al., 2021; Baiyere et al., 2023). DT also reshapes how organisations create and capture value by introducing new business models (Verhoef et al., 2021). Given the criticism that earlier definitions of DT, such as Vial 's (2019), placed emphasis on digital technologies at the expense of organisational change, this thesis draws on the broader definition of digital transformation by Hanelt et al. (2021): 'organisational change that is triggered and shaped by the widespread diffusion of digital technologies' (p. 1160). This definition is well-suited for examining how organisations in natural resource management are shaped by and respond to digitalisation.

DT is undoubtedly a complex process (Amadoru et al., 2026; Holotiuk et al., 2024), often driven by top-down leadership initiatives. The response to seizing the opportunities presented by digital technologies is often undertaken by relatively new C-suite executives such as Chief Digital Officers (CDOs), who take responsibility for leading organisations on their DT journeys (Davison et al., 2023). One way in which CDOs have approached capturing novel value from digital technologies is through digital innovation (Opland et al., 2022). Digital innovation is seen to have transformative power (Yoo et al., 2024) and has been influentially defined by Yoo et al. (2010) as 'the carrying out of new combinations of digital and physical components to produce novel products' (p. 725). As digital technologies are integrated into business processes, organisations begin to explore and exploit digital innovation pathways that lead to DT (Hinings et al., 2018; Hund et al., 2021). Digital innovation is framed as a strategic initiative (Kohli & Melville, 2019) that enables DT when several digital innovation efforts bring about new actor constellations and structures, upending or complementing existing organisational routines and beliefs (Hinings et al., 2018). Although CDOs have been responsible for leading digital innovation in organisations (Tumbas et al., 2020), research has challenged the assumption that innovation agency is centralised (Nambisan et al., 2017). Opland et al. (2022) show that digital innovation in organisations can emerge organically and can be driven by 'ordinary' employees through bottom-up initiatives rather than being confined to dedicated R&D units or departments.

As digital transformation and innovation have evolved, there has been a shift in how data are perceived. Previously implicit in its role in driving organisational and economic change, data have come to occupy a central position (Yoo et al., 2024). Organisations are now increasingly using data to orchestrate algorithms that generate new ways of representing and transforming how the organisations position their digital offerings and carry out their work (Baiyere et al., 2023; Ashrafi et al., 2025). In addition,

effectively mobilising data has been advocated as a means of delivering solutions to achieve global or societal goals such as increased sustainability, extending beyond the organisational boundaries (Davidson et al., 2023). However, data are not simply waiting to be gathered (Clarke, 2019). Yoo et al. (2024) call for greater scrutiny into not only how scholars have treated the role of data in innovation, but also how data are generated, collected, verified, and used. In this thesis, I adopt the view that data are constructed; what gets selected in data collection is shaped by the purposes and perceptions of those doing the collecting, meaning that certain attributes are captured and others are omitted (Clarke, 2019). In the context of natural environments, involving multiple stakeholders with different environmental values, homogenising data may not be collectively agreed upon and risks failing to represent diverse communities and perspectives (Ashrafi et al., 2025). Digitalising natural environments therefore calls for a more careful consideration of how data become meaningful, for whom, and to what effect (Ashrafi et al., 2025).

3. Conceptualising the Capture and Display of Natural Environments

Unlike many of the organisational or administrative phenomena examined in Information Systems (IS) research exploring digitalising natural environments as a phenomenon presents distinct theoretical and practical challenges that require having a clear conceptual foundation (Lumineau et al., 2025). It therefore becomes important to develop a careful understanding of the complexities of the phenomenon. *Phenomena* can be defined as regularities or observed data patterns that challenge existing knowledge and advance scientific discourse (Alvesson & Sandberg, 2024; Von Krogh et al., 2012). Phenomenon-based research builds on this by describing and conceptualising real-world phenomena in ways that advance knowledge and challenge existing assumptions, and offer new and alternative ways of framing empirical problems (Von Krogh et al., 2012; Gkeredakis & Constantinides, 2019; Schwarz & Stensaker, 2016). The distinct conceptual and empirical challenges that will emerge when studying the digitalisation of natural environments therefore require a clear point of departure.

For a phenomenon to be thoroughly investigated, it should first be clearly defined, and its nature and characteristics should be highlighted (Lumineau et al., 2025). This chapter focuses on developing a sensitising device (Klein & Myers, 1999) to establish how the process of digitalising natural environments can be approached analytically. In the following sections, I propose and elaborate on three elements that are key in structuring this process: *objects of digitalisation* (3.1), *data* (3.2), and *digital representations* (3.3). The selection of these elements was informed by Jones (2019) who suggests that we must consider what counts as a phenomenon in relation to real-world events and objects, and whether it is capable of being quantified. In turn, what aspects of a phenomenon can be abstracted and recorded into data depend on the characteristics of the objects involved (Webber et al., 2023). The data must then be represented in formats that make them usable (Jones, 2019). In the sections that follow, I also examine how physicality and the characteristics of natural objects shape what gets abstracted into data, and how different views of data influence what should be considered and represented in the digitalisation of natural environments.

3.1 Objects of Digitalisation

Digitalisation has had far-reaching implications for communities, economies, and industries (Ghawe & Chan, 2022). In many contexts, digital technologies are primarily used to support administrative and coordination activities (Lyytinen, 2022) such as record keeping and communication, largely complementing existing organisational work structures rather than fundamentally reshaping them (Hirschheim et al., 2012). Over the past few decades, as digitalisation has gained a stronger foothold across a wide range of industries, the starting point of what is to be transformed has shifted significantly (Forman et al., 2014). Digital-born companies such as Google and Spotify, are often at the forefront of this shift (Sebastian et al., 2017), expanding their core business catalogues and value propositions, as well as designing their organisational processes around digital technologies (Wessel et al., 2021). Such transformation has often involved streamlining workflows and rethinking how value can be created (Vial, 2019). In these settings, organisations have increasingly innovated on 'malleable intangibles' (Nylén & Holmström, 2015) such as data, information, and digital content (Kallinikos et al., 2013; Yoo et al., 2010) as the core business objects of digitalisation. Intangible objects are particularly receptive to digitalisation as they can easily be standardised and automated.

Beyond these administrative and intangible objects, digitalisation also extends to industries whose core business and key objects of digitalisation are considerably more tangible and physical, such as urban cities, manufacturing, mining, and automotive industries. In these domains, digital technologies have been used to redefine product offerings, improve customer experiences (Svahn et al., 2017), and enhance operational efficiency and safety (Westergren & Holmström, 2012). Despite the growing presence of digital technologies in physically centred industries, their core processes and objects continue to exist in the physical world. As a result, digitalisation in these settings involves an interplay between physical environments and generative digital technologies (e.g., Velsberg et al., 2021). Nevertheless, the role of the physical in shaping digitalisation has often been understudied in IS research. Recent work by Lorenz et al. (2025) echoes the growing calls within IS research for greater attention to physicality, emphasising its indispensable role in shaping consumer experiences and offerings. Although studies acknowledge that digitalisation unfolds in material and physical settings, more knowledge on the ways in which physical objects and their properties can constrain or enable digitalisation is needed. This reflects the challenge in understanding and defining what physical means not just in IS research

but also in other domains. Addressing this gap requires asking a more foundational question about what it means for something to be physical. This question extends beyond IS research, and is one that scholars across different fields, including philosophy have long grappled with.

The notion of something being 'physical' is habitually taken for granted as existing or a 'trivial truth or an obvious falsehood' (Witmer, 2018, p.457), and has been defined using multiple approaches – none without question or controversy. Similarly, defining physicality has not been an easy task in research. Physicality is often ontologically contrasted with non-physical things (Mills, 1997) with Büscher (2006) stating:

Physicality is not the opposite of digitality

This, however, does not define what physicality is and runs the risk of oversimplifying what it entails. Ghazali (2006) therefore proposes that physicality can be looked at as more than mass and can be interpreted based on our understanding of physical attributes such as directness, locality, and visibility. Drawing on these perspectives, in this thesis, physicality can be understood as exploring more than matter, encompassing presence, temporality, spatiality, and location.

Much of IS research sees physicality conflated with or overshadowed by materiality. In IS, physicality has been implicitly recognised, with materiality being predominantly and explicitly researched. Materiality has been defined by Law and Mol (1995), albeit simplistically, as a 'matter of solid matter' by. Leonardi (2013) further defines materiality as 'the arrangement of artifacts' physical and digital materials into particular forms that endure across differences in place and time are important to users' (p. 42). At times, the terms material and physical have been used interchangeably (Stoljar, 2010). This is understandable, given that the doctrines of physicalism and materialism are similar in their grand claims (Tiehen, 2018), where physicalism is related to physical objects (Stoljar, 2001), which are closely linked to material objects and thereby to materialism (Smart, 2026). Leonardi (2010) also argues that physical objects constitute materials with properties that offer either opportunities or constraints. Yet while the importance of materiality in IS research is evident, Robey et al. (2013) argue that physical objects are often treated as cultural symbols or substrates of the digital (Leonardi, 2012). Although Leonardi (2012) states that materiality is not a synonym of physicality, meaning that these two concepts should be treated as different, physicality still remains an underexplored topic in IS research.

Addressing this challenge in a recent literature review, Goebeler et al. (2024) highlight how tangible physical objects in a spatiotemporal context can shape digital innovation. Such tangible objects are characterised by spatial attributes such as location, mass, shape, and volume (Faulkner & Runde, 2019). According to Goebeler et al. (2024), physicality plays four main roles in digital innovation as a *subject*, *vessel*, *context*, and *nexus* of digital innovation. First, when physicality is treated as the *subject* of digital innovation, the physical object is central to what is being transformed by digital technologies. Second, under physicality as a *vessel*, the physical attributes of an object act as key factors in determining how digital innovation can be carried out. In this view, physicality can either enable or constrain innovation by influencing what can be digitalised. Third, when physicality is understood as a *context*, the focus lies on how physical objects interact with their surrounding physical world in empirical settings. This perspective foregrounds underlying assumptions and contextual conditions under which digital innovation unfolds, specifically how and why phenomena occur. Lastly, following Goebeler et al. (2024), physicality as a *nexus* emphasises an entanglement of physical and digital components as well as mutually constitutive relationships between actors, which result in hybrid phenomena.

In their analysis, and in line with much research in IS, Goebeler et al. (2024) direct their attention towards artificial objects, which they describe as human-made objects, as the focus of digitalisation and digital innovation. However, the focus on artificial objects does not fully capture what digitalising natural environments involves, given that natural objects differ fundamentally from artificial ones. That said, there has been a growing body of IS research that extends its empirical scope beyond artificial objects to natural objects. For example, 'ocean informatics' was proposed by Baker et al. (2005) as a way to use [digital] technologies to study, capture, and manage data about oceans and the abundant biodiversity therein. Ocean systems (including plant and aquatic life) are defined by complex processes that react to external stimuli such as wind and sunlight (Baker et al., 2005). As a result, collecting biological data is often more challenging due to the wide range of species across different habitats (Costello & Berghe, 2006). Monteiro and Parmiggiani (2019) similarly examine marine environments where rich flora, fauna and diverse geological formations present distinct challenges for digitalisation.

Outside of IS research, domains such as agriculture and farm management have also explored how technologies such as soil and water sensors can be used for crop monitoring (Cesco et al., 2023; Fountas et

al., 2015). Still, there is scant research that foregrounds the distinct nature of the objects involved. Instead, natural environments often primarily serve as a setting for deploying digital technologies without considering the specific characteristics of the focal object of digitalisation. What then happens is that these characteristics of natural objects[6] remain implicit and under-theorised.

Drawing a hard line in describing the key characteristics of natural objects and their differences from artificial objects in research has not been straightforward. Scholars in philosophy of natural and social systems have debated what really is the difference between such objects. As Vermaas et al. (2011) note, the relationship between the two is better understood as a sliding scale than as a clear-cut distinction (p.11). One way research has approached this relationship is to consider the purpose or human intentionality. Artificial objects are typically designed and produced by human beings with an intended purpose[7] or to realise a specific goal (Baker, 2011; Kroes & Meijers, 2006) and would not exist without individual or human effort. Thus, artificial objects are correlated with the cognitive capabilities of the designer (Wallace, 1996). Natural objects on the other hand are not a product of design or intention (Vogel, 2003) and already exist in nature (Stuermer et al., 2017).

Compared with artificial objects, natural objects are also seen to possess agency that is not as a result of human intention. Moreno and Etxeberria (2005) argue that [in natural objects] agency develops through evolutionary processes that give rise to autonomy. From this perspective, natural objects are deemed autonomous, where they are able to act on their own behalf without the direct intervention of humans (Misselhorn, 2015) and regulate aspects of their environments that are crucial for their existence (Moreno & Mossio, 2015). For example, Gabrys et al. (2025) argue that forests interact with many different ecosystems and biological life forms. Additionally, Sultan et al. (2022) describe agency in terms of [natural objects'] ability to take part in their own development, maintenance, and functioning. This highlights how natural objects are not passive, but rather active and dynamic.

Given the dynamic and evolutionary nature of natural objects, important questions arise about what aspects of them can be captured and quantified (Jones, 2019). These questions point to the importance of

[6] Natural objects are objects that exist regardless of human practices (Baker, 2008).
[7] Here, I am not disputing that artificial objects may be used in unintended ways or produce consequences beyond those anticipated by their designers. What I wish to highlight is how these objects are often created with specific purposes in mind.

exploring how data about natural environments are constructed and understood, which the following section addresses by examining different perspectives on data.

3.2 Data

Data have always been central in the IS discourse (Langefors, 1973; Østerlie & Monteiro, 2020; Aaltonen et al., 2023; Jones, 2019) but have until recently been treated as implicit in supporting organisational operations and tasks such as automating routines. At a rudimentary level, data were described as simple facts or records stored in a computer which could be structured into information (Buckland, 1991). Often, data and information have been described interchangeably, creating some ambiguity in distinguishing between these concepts. This ambiguity is not without reason, as research has historically argued that there exists a relationship between data, information, and knowledge[8] (Stuermer et al., 2017). While Tushman and Nadler, (1978) describe information as relevant and accurate data, Tuomi (1999, p. 7) argues that the reverse can be at hand, with data not being a raw material but rather emerging when value is added to information by putting it into a processable structure. Here, Tuomi (1999) criticises the conventional view that exists on the sequential relationship between data, information, and knowledge, and calls for a reversed hierarchy that foregrounds how data can emerge as a product of cognitive artifacts. Similarly, although information and knowledge are also often used synonymously, Nonaka (1994) highlights that knowledge is information grounded in the beliefs of the agent or holder. Tuomi's (1999) observations on the prevailing assumptions about the notable 'data to information to knowledge' hierarchy were echoed by Alavi and Leidner (2001), who argue that what distinguishes information and knowledge is not the content or structure, but rather the cognitive processing that converts information to knowledge in the mind.

Zins (2007) defines data in two domains: subjective and universal. In the subjective domain, data are simply stimuli we perceive through our senses. In the universal domain, data are signs that represent empirical knowledge. Likewise, Ackoff (1989, p. 1) describes data as 'symbols that represent the properties of events and objects'. Kitchin (2022, p. 6), however, highlights that data can be understood as '…socially constructed, as having materiality, as ideologically loaded, as a commodity to be traded and/or as constituting a public good'. Ackoff's account assumes a neutral, objective understanding of data, in contrast to Kitchin's view, which

[8] My aim here is not to engage in a historical scrutiny of the data-information-knowledge hierarchy but to acknowledge the relationship that exists between these terms which gave rise to the ambiguity in definitions and use in the IS field

positions data as socially and ideologically situated. Although these two views sit at opposite ends of the conversation, advancing the discussion on the role of data in organisations requires attention to a middle ground. Though variations in how data are defined and categorised in IS literature are rife (Jones, 2019), Jarvenpaa and Markus (2020) in their study on data sourcing and partnerships offer three perspectives that help scrutinise how data in organisations are conceptualised and understood (for an overview, see Table 1).

Table 1. *Three key views of data (Jarvenpaa & Markus, 2020)*

Views of data	Focus	Examples of studies
Commodity view	• Data as raw and unorganised • Tokenised perspective • Objective and faithful representations • Data as tradable liquid assets	(Xu et al., 2023), Parvinen et al. (2020), Najjar and Kettinger (2013), Ofulue and Benyoucef (2024), Zhang et al. (2023), Aaltonen et al. (2021) Faroukhi et al. (2020), Alaimo et al. (2020) Wixom et al. (2021), (Xu et al., 2023)
Processual view	• Challenges objectivity of data • Data practices shape reality • Recontextualisation affects objectivity • Data representations are fallible	Jones (2019), Frické (2015), Kitchin (2022), Aaltonen and Stelmaszak (2024), Aaltonen and Tempini (2014), Paparova (2023b), Günther et al. (2022) Alaimo and Kallinikos (2022). Abbasi et al. (2016)
Relational view	• Data meaning depends on actor and purpose • Situated in concrete contexts • Transferability is context-dependent • New organisational arrangements	Leonelli (2015), Alaimo et al. (2020), Aaltonen et al. (2023), Xu et al. (2023), Jarvenpaa and Essén (2023), Jagals and Karger (2021), Guggenberger et al. (2025), Paparova et al. (2023)

First, under the *commodity view*, data are understood through a token lens and interpreted as raw, unorganised facts that symbolically relate to

objects or events. Within this view, data function as sign tokens that represent reality or measure phenomena and are assumed to objectively and faithfully reflect real-world facts. This perspective positions data as not only inputs that provide administrative support, but also as resources for organisational knowing (Alaimo & Kallinikos, 2022; 2024). Organisations seek to identify ways to capture and extract value from data, conceived as a form of raw material, for monetary benefit (Parvinen et al., 2020). In such situations, data can be used to make more informed decisions and improve service or product offerings, operational efficiency, and firm competitiveness (Najjar & Kettinger, 2013; Ofulue & Benyoucef, 2024; Zhang et al., 2023). Accordingly, data have been treated as contextual assets that can be exploited to create value (Vassilakopoulou et al., 2019; Parvinen at al., 2020). This gives rise to organisations monetising data, by either explicitly selling data or wrapping and enhancing products (Wixom et al., 2021; Faroukhi et al., 2020). From such a value-adding perspective, data may also be understood as a tool, a capability, or a form of algorithmic intelligence (Xu et al., 2023). Research suggests that as a commodity, data is a tradable good that can be exchanged on a market (Aaltonen et al., 2021), and this is attributed to the properties of data and the possibilities they offer (Xu et al., 2023). Data are considered liquid assets (Wixom et al., 2021) and can be edited, reprogrammed, aggregated, and repurposed (Alaimo, 2022; Monteiro & Parmiggiani, 2019).

Second, the *processual view* confutes the claims of data objectively representing the world. Under this view, how data are produced, actualised, and reused has implications for the representations of reality (Jones, 2019). The different ways data may be recontextualised impact objectivity. Hence, data may not accurately represent reality and as such are fallible (Frické, 2015). Framing data as benign, objective and pre-existing entities waiting to be collected has been widely criticised (Gitelman & Jackson, 2013). Scholars argue that data should rather be understood as 'epistemological units', generated through techniques and thought systems that are grounded in philosophical beliefs and assumptions (Jones, 2019; Kitchin, 2022). Jones (2019) offers a broader view by showing that data are not a given or things that exist 'out-there' but come to be through the evolving and subjective judgements of individuals. From this perspective, data are not fixed representations of reality but are increasingly recognised as generative and open-ended (Aaltonen et al., 2021; 2023). This open-endedness renders data unbounded, enabling them to be explored in different ways and across different contexts (Aaltonen & Tempini, 2014).

In the digitalisation of natural environments, data design becomes particularly critical. Data design refers to the activities and choices through which specific entities and their corresponding characteristics are recorded and forged in organisational reality as data (Aaltonen & Stelmaszak, 2024). The choices in data design (what gets recorded and how) have subsequent implications for how data are governed and used (Davidson et al., 2023). As Abbasi et al. (2016) argue, evolving data practices create the need to reassess assumptions and questions related to data and their organisational implications. Consequently, scholars increasingly question whether traditional modes or frameworks of data governance are sufficient for addressing the dynamic nature of data (Paparova, 2023; Günther et al., 2022). As data accrue and are diffused across projects and through organisational networks (Parmiggiani et al., 2023), managing the resulting uncertain data flows will require the mindful application of appropriate data governance techniques (Janssen et al., 2020). Within this broader debate, one dominant stream of research continues to define data governance in relation to *data quality management* focusing on dimensions such as *accuracy*, *timeliness*, *completeness* (Janssen et al., 2020) and how data can remain compliant with legal provisions and prevent risks (Benfeldt et al., 2020).

Lastly, under the *relational view*, data are understood in relation to concrete contexts or situations. Jarvenpaa and Markus (2020) argue that relational perspectives on data remain largely implicit within IS research, in contrast to the commodity and processual views of data,. From a relational perspective, data depend on who uses them, how they are used, and for what purpose. Data are viewed in terms of their prospective usefulness as evidence for a phenomenon under study, and the transferability of data across individuals and settings is also considered under this view. Accordingly, the scope of data extends beyond their mere existence towards the conditions under which they are used for different purposes (Aaltonen et al., 2023). Some IS scholars contend that the nature of digital data has evolved and data should now be viewed as 'artifacts' that act as vessels for sense-making rather than simply as 'facts' (Alaimo & Kallinikos, 2022; Aaltonen et al., 2021; 2023). In this regard, data become organisationally embedded, reflecting an interplay between social context and data practices (Xu et al., 2023). A relational view of data therefore highlights the importance of considering the multiple interests that shape how data are embedded, translated, and used (Jarvenpaa & Essén, 2023).

Data have also been described as technological objects where the capacity to represent phenomena lies in their ability to be dematerialised or

liquefied (Aaltonen & Penttinen, 2021; Lusch & Nambisan, 2015) from material artifacts. This liquefaction allows data to be transferred across different contexts (Günther et al., 2022). Building on this understanding, Leonelli (2014) describes data as going through data journeys consisting of successive stages of *decontextualisation*, *recontextualisation*, and *reuse* before they become valuable inputs. First, *decontextualisation* involves ensuring that data are formatted in ways that make them compatible with data from other sources (Galliers et al., 2017). Once data have been processed into compatible formats, they undergo *recontextualisation* whereby they can be adopted and integrated into various contexts. Finally, data are then *reused* to uncover different correlations and new insights, and to serve a purpose and generate value for users.

In the relational view, new organisational arrangements may emerge to support data as they become context-situated (Jarvenpaa & Markus, 2020). These arrangements may arise within or beyond organisational boundaries. Therefore, approaches to orchestrating and governing data need to be adapted as actors and resources are distributed across organisational contexts (Abraham et al., 2019). Within this context, Zhang et al. (2022) identify four critical elements of data governance: data collaboration, activating perceptions for data potential, developing data-related capabilities, and establishing data legitimacy. Data governance spans both intra-organisational and inter-organisational domains (Abraham et al., 2019; van den Broek & van Veenstra, 2015). Collaborating around data in a single firm as opposed to between multiple firms will result in different structures, participants, and processes (Jagals & Karger, 2021). As such at the interorganisational level, collaborations around data may take the form of data spaces to address the complexities of using and sharing data (Guggenberger et al., 2025; Paparova et al., 2023).

Although each view offers a distinct perspective on data, they are not mutually exclusive and are likely to co-exist within the same digitalisation efforts, depending on the goals of those involved. In the context of digitalising natural environments and where multiple stakeholders hold different values, the view of data that prevails will shape what gets recorded about the phenomenon and used. For data about natural environments to be meaningful to different stakeholders, they typically require visual representations. As Gitelman and Jackson (2013) argue, it is through these representations that data acquire their rhetorical function and become a basis for explanation. Digital representations are

therefore central to how data about natural environments are understood and used.

3.3 Digital Representations

Digital representations have become an essential concept in Information Systems (IS) research, reflecting how digital technologies model, mediate, and shape real-world phenomena (Recker et al., 2021). Digital technologies enable the digitisation of physical and digital phenomena (Bailey et al., 2012). Through algorithmic manipulation of data, digital technologies can provide a rendition of these phenomena (Østerlie & Monteiro, 2020). At their core, digital representations consist of data that are 'syntactically and semantically' structured to depict reality to different users or groups of people (Fossum & Kempton, 2020, p. 6). Bailey et al. (2012, p. 1486) define a [digital] representation as:

> *something that stands for something else within the bounds of a particular speech community, culture, or community of practice…and also point[s] to something definite in the world, a physical entity*.

Such a definition highlights that representations are selected and constructed (Ciborra, 2006; Fossum & Kempton, 2020) and obtain meaningfulness and representational status when embedded in systems or infrastructure that process any acquired information (Müller, 2008). Digital representations being constructed, particularly about natural environments, points to the ways in which choices and techniques to generate and interpret data (Mikalsen & Monteiro, 2021) can admit some perspectives and not others (Gabrys, 2016; Jones, 2019). Representations of phenomena are partial; they are never fully accurate, given that contextual information and subjective nuances may be omitted (Grudin, 2001). Despite this, digital representations are seen as 'faithful' (Burton-Jones & Grange, 2013) to the extent they closely relate to the corresponding physical referent. Here, digital representations are *like* reality and can substitute for aspects of the real world (Hovorka & Mueller, 2025). This perceived fidelity is gained through continual quality checks and correctness of data (Østerlie & Monteiro, 2020), highlighting the dynamic nature of representations, not as static models but as constantly evolving constructs that require continuous refining. If digital representations are to 'faithfully' represent given phenomenon, what data will be collected and shared to achieve this, and whose interests will be prioritised in the process (Ciborra, 2006)? These questions are crucial for unpacking how natural environments can be digitalised.

The notion of digital representations in IS research is overall grounded in representational theory, which posits that information systems are designed to provide and depict a real-world phenomenon as perceived by specific user groups (Wand et al., 1995). Here, information systems capture and convey meaning based on a user's interpretation and understanding of reality – hence, representations are non-neutral or non-universal and rather reflect the mental models and assumptions of designers and users (Wand et al., 1995). According to Burton-Jones and Grange (2013) representation theory suggests that information systems have three components: representations, physical structure, and surface structure. These representations are interacted with and accessed through the physical and structure layers with the goal to create representations that closely reflect the real-world aspects meant to be modelled. This aligns closely with the practice of conceptual modelling which involves developing and using representations to capture the attributes of both static and dynamic phenomena in the real-world systems (Burton-Jones & Meso, 2006; Recker et al., 2021). These conceptual models encompass scripts based on combining attributes and rules within a given context, which shapes how the models are created and applied (Recker et al., 2021).

Traditionally, the role of information systems has been understood to be representing and mapping physical domain. However, Recker et al. (2021) challenge such assumptions and recognise the role digital technologies play in shaping and governing reality. Although digital representations never fully replicate the exact nature of the physical phenomena they model (Monteiro & Parmiggiani, 2019), they play a role in [digital] mediating – how IT facilitates and shapes the process of capturing and bridging the multiple realities of physical and digital (see Jonsson et al., 2018; Recker et al., 2021). In the context of natural environments, digital representations mediate interactions with nature that in ways that are not feasible in the real-world (Webber et al., 2023). This capacity is afforded by technologies' ability to create and shape our physical environments (Baskerville et al., 2020). From this perspective, digital representations are not passive; rather they are performative, embedded in contexts and actively influence them (Østerlie et al., 2012). For example, in the context of offshore oil and gas production, Østerlie and Monteiro (2020) investigate the efforts to replace organisational practices of sand monitoring with digital representations and highlight how the physical and digital were not dichotomised but rather interacted to create digital representations in organisational action, suggesting that representations are socially and materially embedded.

To deepen our understanding of digital representations, it is important to consider how they are categorised semiotically. Bailey et al. (2012) draw on C.S. Peirce's[9] taxonomy of indices, icons, and symbols to describe representations. First, as indices, representations directly point to or mirror their physical counterparts based on an existential relationship (Bailey et al., 2012). Second, as icons, representations closely resemble the physical entity being signified, moving beyond merely mirroring (Østerlie & Monteiro, 2020). Third, as symbols, representations are interpreted as abstract entities in simulations and have the ability to conduct and exemplify scenarios when applied in social processes (Zammuto et al., 2007). In digitalising natural environments, Webber et al. (2023) highlight that digital representations can take the form of visualisations (graphs) or naturalistic environments such as virtual immersive environments. The capability of representations to be algorithmically manipulated and transformed in this way is based on their ability to be liquefied – decoupled from physical objects (Monteiro & Parmiggiani, 2019). For instance, simulations are being explored in agricultural contexts for precision farming, to accurately monitor livestock and plants (Norton et al., 2019; Pylianidis et al., 2021). Such application shows how digital representations are integral in shaping natural and evolving environments. The same logic can be used in other natural environments such as forests, as trees are biological organisms whose growth is determined by biological rules. Digital representations like point clouds can be used to determine the range of forest fire hazards or to plan forest management scenarios (Murtiyoso et al., 2024). These representations also offer stakeholders a way to visualise and better understand why decisions about forests are made (Murtiyoso et al., 2024).

In this chapter, I have examined three key elements (objects, data, digital representations) that form the basis of a sensitising device for studying the digitalisation of natural environments. In the next chapter, I detail how I studied the Swedish forest industry as a context for digitalising natural environments and the methodological choices that informed this investigation.

[9] Peirce, C. S. (1932). *Elements of Logic: Collected Papers of Charles Sanders Peirce*. Vol. 2. Harvard University Press, Cambridge, MA.

4. Methodology

This chapter provides an overview of the overarching research approach I have used in my thesis and the appended papers. First, I discuss the philosophical foundations that informed this research. Following this, I describe the qualitative case study approach used in Papers 1 – 3 and the design science research approach used in Paper 4. I then describe the individual cases in each paper and detail the data collection and analyses used. Lastly, I reflect on my role as a researcher and address the ethical considerations that guided my research.

4.1 Philosophical Foundations

This thesis is grounded in philosophical assumptions that recognise the complex, socially constructed nature of reality and knowledge. According to Najjar and Kettinger (2013), researchers approach a subject using either implicit or explicit assumptions about the world, which can be ontological in nature. Ontology concerns 'what is' regarding the very essence and nature of existence (Crotty, 1998; Smith, 2003). Researchers are often faced with questions about what entities exist or can exist and whether the reality to be investigated is objective and external to the individual (Bricker, 2016; Burrell & Morgan, 1979). Ontological assumptions can be analysed using two strands of debate: nominalism and realism (Burrell & Morgan, 1979). The nominalist position argues that the social world is made up of names, concepts and labels that are used to structure reality, and there being an absence of a 'real structure'. Realism, however, argues that the world consists of tangible and fixed structures and that the social world exists independently of the individual's recognition of it (Burrell & Morgan, 1979, p.4). Similarly, the opposing schools of thought on realism and idealism have been applied to describe reality in philosophical debates. Realism postulates that the physical world exists outside of human perception, whereas idealism claims that reality is in a sense shaped by humans' conscious actions (Okasha, 2016).

In my thesis, I draw on Chowdhury's (2019) categories of ontological positions – objectivism and constructivism to demarcate the assumptions that ground my research. I position my research under constructivism, where Chowdhury (2019) highlights that within this ontological position, reality is not fixed and can be perceived in different ways. Specifically, I ascribe to the social constructivist philosophy, which assumes that people engage and make sense of the world based on social and historical

perspectives. As a result, people create meaning from interpreting the world they are engaged in (Crotty, 1998). A researcher using this philosophical lens looks for complexity of views (Creswell, 2009). In the context of this thesis, the social constructivist lens is particularly important for understanding how natural objects are represented. Data, often perceived as objective and factual (Ackoff, 1989), are subject to interpretation which will shape how digital representations depict the physical and social world.

Ontology and epistemology are related. Epistemological assumptions explore how we can begin to know and understand reality (Burrell & Morgan, 1979; Chowdhury, 2019). In this thesis, I use the epistemological positions of positivism and anti-positivism to give a binary structure to my argumentation. I therefore position my research under the umbrella of anti-positivism. Anti-positivism here relates to the broad term of relativism which, as described by van de Ven (2007), served as a set of alternative philosophies in denial of positivism. This distinction between positivism and relativism is done on the basis of positivist assumptions and beliefs grounded in scientific methodology to develop an unmediated and objective truth of reality (van de Ven, 2007). Such assumptions are not to be inferred from the research I conducted in this thesis. However, the broad assumptions of relativism – that reality is socially constructed and is based on the interests and viewpoints of scientists' interactions with others – is salient in the thesis.

Using Morgan's (1980) identification of paradigms as a means to explain views of social reality, I situate my research under the interpretivist school of thought. The interpretivist paradigm, as explained by Morgan (1980), entails understanding society from the standpoint of the participants involved and in action. This is corroborated by my aim to understand how data in their capacity to represent natural objects can contribute to transformative outcomes. In interpretivist research, data are seen as personal understanding and constructions of other people's constructions (Walsham, 2006). In interpretive studies, researchers assume that access to reality is only through social constructions such as language, consciousness, shared meanings, and instruments, and the qualitative interpretivist research methods urge researchers to engage with people in real organisations (Myers, 2020). Interpretative research is often used to describe a situation, problem, phenomenon, or provide information about or discuss attitudes towards the issue under investigation (Myers, 2020). The interpretivist perspective therefore aids identification and gaining a deeper understanding of different actors' thoughts and approaches to representing natural objects and digitalising natural environments.

Often, philosophical assumptions or stances are regarded as dualistic. The categories of positivism and interpretivism are not exhaustive as proposed by Chowdhury (2019), therefore my research also falls in line with the pragmatist epistemology. Pragmatism focuses on concrete real-world issues and provides the boundaries for a pluralist understanding of multiple truths (Kelly & Cordeiro, 2020). The pragmatic philosophy is also suitable for understanding the connections between knowledge and action in a specific context (Kelly & Cordeiro, 2020). My role as a researcher in the thesis tends towards that of an engaged scholar where I collaborate with practitioners to study a problem in real-life situations (Fisher, 2010).

In IS research, there has been an issue of placing more emphasis on rigour in research than on relevance to wider societal narratives (Benbasat & Zmud, 1999). This discrepancy between rigour and relevance, according to Benbasat and Zmud (1999), lies in multiple factors. There has been overemphasis on rigour by researchers as much as a lack of cumulative tradition. The IS field was overcome with a plenitude of theoretical frameworks such that it was difficult to create streams of research (Benbasat & Zmud, 1999). Again, much precedence was given to abstracting research and focusing on methodological precision and accuracy. Conforming to domain norms and values was highly important in ensuring that academic rigour was attained (Benbasat & Zmud, 1999). With much attention on rigour, relevance for industry practitioners was found lacking (Davenport & Markus, 1999; Benbasat & Zmud, 1999). Consequently, there needs to be a balance between research and real-world problems and its usefulness for practice.

In an attempt to highlight the need for relevance in research without forgoing rigour, Mathiassen (2002) adopts the term collaborative practice research to highlight the combination of action research, experiments, and practice studies. Throughout my thesis process, I mainly embodied a role where I was allowed access to gather data (Fisher, 2010), but stopped short at identifying as an action researcher, where the aim is to develop solutions to the practical problems of the organisations or stakeholders being studied (Mathiassen et al., 2012)

These assumptions had direct implications for my methodology of choice, as each one had consequences for how knowledge could be investigated and obtained (Burrell & Morgan, 1979). By subscribing to the view of social reality focused on the subjective experiences of individuals, an ideographic approach allows for researchers to understand what is unique and particular about phenomena (Burrell & Morgan, 1979). Aiming to understand the different and complex perspectives surrounding data and

the implications they can have in digitalising natural environments meant that a qualitative approach was most suitable. A qualitative approach in my thesis allowed me to engage with and understand phenomena emerging in both the digital and the physical worlds (Cecez-Kecmanovic et al., 2020).

4.2 Qualitative Case Studies

Qualitative studies address aspects of social life such as interest and social backgrounds (Holliday, 2016) and enable researchers to gather data not only about things that *are* happening, but also about things that *have* happened, as well as peoples' thoughts and how they talk about a topic (Corbin & Strauss, 2015). Whereas quantitative methods, grounded in the assumptions of the positivist perspective, seek to operationalise constructs using measurement variables (Straub et al., 2022) and test pre-specified hypotheses (Kaplan & Duchon, 1988), qualitative methods explores answers to questions through the study of social settings and their inhabitants (Berg, 1998). Qualitative research allows researchers to understand the context within which decisions and actions occur (Myers, 2020). Additionally, through qualitative studies:

> *…we can explore a wide array of dimensions of the social world, including the texture and weave of everyday life, the understandings, experiences and imaginings of our research participants, the ways that social processes, institutions, discourses or relationships work, and the significance of the meanings that they generate. We can do all of this qualitatively by using methodologies that celebrate richness, depth, nuance, context, multi-dimensionality and complexity rather than being embarrassed or inconvenienced by them. Instead of editing these elements out in search of the general picture or the average, qualitative research factors them directly into its analyses and explanations. This means that it has an unrivalled capacity to constitute compelling arguments about how things work in particular contexts.* (Mason, 2002, p. 1)

In Papers 1-3 of this thesis, I use of qualitative case studies to explore various phenomena in each paper. Case studies are particularly helpful for researching 'practice-based problems where the experiences of actors are important and the context of action is critical' (Benbasat et al., 1987 p. 369). Case studies allow for deeper understanding of dynamic settings (Mason, 2002) when studying phenomena such as the digitalisation of

natural environments. Using case studies means that the phenomenon to be studied will be positioned in a specific context.

Case studies are valuable as they allow researchers to capture reality with descriptions and greater detail (Galliers, 1991). These techniques were important in my approach to my individual papers, where giving descriptions of the settings and contexts aided understanding of the phenomena to be studied. However, when adopting a qualitative approach to research, it is however not uncommon to receive criticisms and questions about the potential to generalise a study. As Galliers (1991) points out, case studies' weaknesses are typically the result of being limited to a single organisation, making generalisations difficult. However, as Lee and Baskerville (2003) highlight, empirical data from case studies can be generalised to theoretical statements which can offer contributions to the overarching domain of study. In terms of contribution to theory, interpretive research is not focused on falsifying theory but rather uses theory as a lens through which the world can be viewed (Klein & Myers, 1999). Data from case studies can also be abstracted and generalised into concepts and theories or offer rich insights to a phenomenon (Lee & Baskerville, 2003). Mason (2002) suggests that qualitative research should not be idiosyncratic but rather focus on producing generalisable arguments. Therefore, although Papers 1 to 3 describe case-based studies, I aimed for my arguments to be generalisable to other contexts and phenomena.

Furthermore, with my approach to the thesis being primarily performing qualitative case studies, an interpretivist perspective guided how data were collected, which was mainly through interviews. According to Walsham (2006), interviews are important as they serve as a crucial way of gaining accessing to the interpretations and perspectives of informants included in a case or field study. Interviews were complemented with secondary documents in the form of internal reports, published annual and project reports, and web pages. Regarding interpretive and qualitative research, Walsham (2006) caution that data and findings are susceptible to biases, prejudices, and knowledge of researchers. Considering Klein's and Myers' (1999) principle of interaction between the researcher and the subjects (p. 81), I critically reflect in later sections of this chapter on how my preconceptions about the context influenced the construction and organisation of the research data. It is also worth noting that within an interpretivist perspective, respondents describe their interpretations of events and situations (Walsham, 1995) shaped by their interactions with the researcher. By adopting an interpretivist stance, there needs to be a critical reflection on the social, cultural, and

historical background of the research setting, to give the intended audience insights into the context of the research (Klein & Myers, 1999). In this thesis, I do this by describing the state of Swedish forestry and the historical, cultural, and political aspects that are present in my research setting.

4.3 Design Science Research

The use of methodological pluralism to investigate relevant phenomena has been appropriate for the aims of this thesis. Design science research (DSR) fundamentally focuses on solving a problem by creating innovations that support and explain the analysis, design, implementation, and use of information systems (Hevner et al., 2004). According to Gregor & Hevner (2013, p. 337), DSR involves constructing socio-technical [IT] artifacts, which entail models, constructs, and methods (Hevner et al., 2004), that serve as a basis for both evaluation and interventions in information systems. Hevner et al. (2004, p. 82–83) propose seven guidelines, although not mandatory, to assist researchers in understanding the requirements for design science research: (1) DSR must produce a viable artifact, (2) and this artifact should be developed for a specified problem. (3) To effectively solve the problem, the artifact needs to be evaluated, to assess its use and quality. Additionally, (4) DSR must provide novel contributions such that the artifact developed either addresses an unsolved problem or provides a more effective way for solving an existing problem, (5) and will require rigorous methods to do so. (6) The means to design an artifact should entail a search process and (7) the results must be effectively communicated.

In the context of design science research within the IS discipline, theories are often used to provide prescriptions on how artifacts can be designed and developed to support achieving a specific goal (Jones & Gregor, 2007). As Walls et al. (1992) explain, 'a design theory is a prescriptive theory based on theoretical underpinnings which says how a design process can be carried out in a way which is both effective and feasible' (p. 37). In Paper 4, the emphasis is on developing a design theory that provides prescriptive guidance for an IS artifact. The development of a design theory in Paper 4 aims to contribute to Gregor's (2006) Type V theory for design and action. According to Gregor (2006), the theory for design and action shows how to do something and includes prescriptions for building applications. Thus, the focus of the paper leans more towards theoretical development than constructing the artifact itself (Kuechler & Vaishnavi, 2008). In doing so, the study draws on a kernel theory – descriptive theories that govern the design of an artifact (Gregor &

Hevner, 2013; Walls et al., 1992) – as a foundational element of the design theory presented in the paper.

4.4 Swedish Forestry as Investigative Context

This thesis examines the digitalisation of natural environments within the empirical setting of the Swedish forest industry. Swedish forestry offers a unique context to study this thesis' phenomenon. Sweden is an extensively forested country with a large and historically significant forestry sector (Lindahl et al., 2017), where exports generate around €17 billion annually[10]. For over a century (Fridman & Danell, 2024), forestry has been central to Sweden's economic identity. This position was built over centuries of economic and industrial development. The industrialisation of Swedish forestry, which was driven by international demand for timber, transformed forests into a commodity for exportation. From industrialisation through to mechanisation, successive waves of technological change ushered in an expansion of the industry and its adjacent sectors such as sawmill and pulp industries (Antonson, 2011). However, this focus on industrialisation was prioritised over other uses of the forests, such as reindeer husbandry and hunting (Sandström et al., 2011) which created tensions between different stakeholder groups such as local communities, and indigenous groups. These tensions between economic and ecological or social interests remain noticeable today (Pokorny-Kindlman, 2024; Sandström et al., 2016).

Currently, digital technologies are increasingly disrupting the traditional value chain and redefining how forestry actors collaborate (Holmström, 2020, Koukouvinou et al., 2023). Swedish forestry has evolved over the years and is actively exploring how data can be used to develop digital solutions that not only improve commercial goals but also address ecological and social sustainability challenges. Central to this is the capacity of digital technologies to separate data from its material forms (Aaltonen et al., 2023), giving rise to new synthetic representations (Monteiro & Parmiggiani, 2019) of natural environments. Understanding this history is important because it helps explain why digitalisation in Swedish forestry is a complex and nonneutral process that takes place within a landscape already shaped by competing interests, contested values, and power relations.

[10] Skogsindustrierna 2025 Snapshot of the Swedish forest industry

In what follows, I offer a brief overview of the history of the Swedish forest industry, the stakeholders involved, the sustainability pressures it faces, and its use of digital technologies in the industry.

4.4.1 The Development of Swedish Forestry

The Swedish forest industry has not remained stagnant; it has evolved continuously, albeit arguably at a slower pace than other industries. From the mid-nineteenth century (Mårald & Westholm, 2016), Sweden underwent many changes which saw a transition from a mainly agrarian lifestyle to a more industrial one (Sylwan, 2011). Forestry was seen as an industry capable of facilitating Sweden's shift into a modern and industrialised country having established itself as the leading industry in export revenues through sawmilling (Kardell, 2016). Much like other aspects of society, forestry was revolutionised to mirror social changes such as mechanisation, industrialisation, and capitalisation (Josefsson & Östlund, 2011). Working methods changed and knowledge on the use of natural resources evolved (Antonson, 2011) to match international competition for export markets (Persson, 2011). For much of its early history, Swedish forestry was a low-technology, labour-intensive industry that relied on manual work and simple hand tools (Lundgren, 2011; Josefsson & Östlund, 2011). This way of working remained relatively unchanged for decades, until the mid-twentieth century, with only minor technological developments happening. However, that began to change in 1930s, when forestry companies started exploring ways to improve their working methods (Persson, 2011). The introduction of motorised equipment such as chainsaws in the 1950s marked a pivotal turning point in forestry, with daily productivity increasingly greatly (Lundgren, 2011). Chainsaws replaced traditional saws and axes for felling trees, and tractors replaced horses for hauling timber (Skogsstyrelsen, 2020). The pace of change accelerated in the industry and large machines such as harvesters were introduced to perform several felling operations simultaneously (Persson, 2011), with little attention being paid to forest ecosystems (KSLA, 2024).

Technological developments were initially driven to remain competitive, but reducing harvesting and transport costs also became important for maintaining profitability (KSLA, 2024). However, the predominant focus on timber production had large ecological consequences for the forests (Östlund & Zackrisson, 2000). The intensity of industrial forestry practices faced growing opposition from environmental groups which led to concrete changes such as the reduced use of herbicides (KSLA, 2024). Criticisms against selective cutting and forest grazing also emerged

during the mid-twentieth century, which eventually resulted in new forest management systems (Skogsstyrelsen, 2020; Östlund et al., 1997). The Swedish forest industry also made investments in replanting after harvesting (Holmen, 2025). Forest management in Sweden evolved considerably from largely unregulated forest use, through legislative restrictions to protect timber growth, to a more complex large-scale market-oriented regulations (Gaunitz, 2000). In this way, Swedish forestry shifted from a labour-intensive industry that relied on manual work to a capital-intensive and technologically advanced one (Josefsson & Östlund, 2011), with significant consequences for both the industry and the different stakeholders and communities that depended on it.

4.4.2 Stakeholders and Competing Interests

Sweden is a heavily forested country, with forest ownership is distributed across various actors. Individual owners hold the largest share, controlling 47.7% of forested land, followed by private limited companies at 24.3%. The remainder is held by public owners, including the state at 20.4%, the church at 6.1% and the municipalities at 1.5% (Skogskunskap, 2025). This distribution of ownership highlights the central role that forests play in the lives of individuals across Swedish society, meaning that decisions about how forests are used and managed reflect a broad range of interests and priorities. Lindahl and Westholm (2012) describe the main actors in Swedish forestry as being forest owners, forest industry companies, environmental NGOs, the state, and the indigenous Sámi people.

Beyond ownership, forests hold historical and cultural significance in Swedish society. Forests have non-material uses that extend well beyond economic profit, as they have long featured in folklore and myths and hold spiritual and aesthetic values that have shaped Swedish cultural life (Skogsstyrelsen, 2020). This cultural connection is also reflected in how Swedes interact with forests in their everyday lives. Although fewer Swedes are working directly in forestry than before, many maintain a strong connection to the forest through leisure and recreational activities (Antonson, 2011). This connection is further supported by the fact that it is not only landowners who have access to the forest. Thanks to ‘allemansrätten’, the Right of Public Access, everyone is provided the privilege of walking, camping, or picking mushrooms or berries in the forest, as long as they do so responsibly and with respect for wildlife, nature, and other people (KSLA, 2024).

For the Sámi people, the relationship with the forest carries deep cultural and economic significance. Historically, forests have served as a source of subsistence and trade in the semi-nomadic Sámi livelihood (Skogsstyrelsen, 2020) and they are still used for hunting and reindeer husbandry (Naturskyddsföreningens, 2020). These uses are also recognised under law, giving the Sámi the right to let their reindeer graze within the reindeer management zone which encompasses 50% of Swedish land (cf. Sandström et al., 2016). Many of these practices prevail in Sámi culture, making forests not just an economic resource but a central to their identity and way of life.

The varied uses and relationships with the forest mean that Swedish forestry has long been a site of competing interests and perspectives (Sandström et al., 2016). Forests have multiple uses, so conflicts often centre around forest degradation, conservation priorities, and rights to land and resources (Eckerberg & Sandström, 2013). Balancing these different interests and values has become increasingly difficult as sustainability concerns have grown in prominence in recent years.

4.4.3 Focus on Sustainability in Swedish Forestry

Sustainability concerns in Swedish forestry are nothing new. As far back as during drafting of the 1903 Forestry Act, the need for sustainable timber production was recognised. Replanting after harvesting was made a legal requirement to address general societal concerns about the environment (Nylund, 2009). For much of the twentieth century, forest policies were primarily focused on the wellbeing and interests of farmer communities and industrial workers with the overarching goal of maximising production and ensuring forest lands were being responsibly used to provide sustainable yield (Antonson, 2011; Nylund, 2009).

However, from the 1970s onwards, a shift occurred. Environmental awareness became cemented not only in public opinion but also in legislation driven by the growing criticism of large-scale clear cutting and the use of pesticides (Antonson, 2011; Skogsstyrelsen, 2025). These criticisms continued until the 1993 revision of the Forestry Act where production and biodiversity objectives became equally balanced (Heder Brandt et al., 2023). Under this revised framework forest owners were free to manage their forests as they saw fit provided it was within the bounds of sustainable forest management. This iteration of the Forestry Act also allowed for a diversification of forest management practices such as alternative silvicultural methods (Skogsstyrelsen, 2020).

Despite this shift, Swedish forestry finds itself a crossroads where tensions between economic and ecological priorities still remain, inciting debates on nature conservation and use of the forest[11] (Heder Brandt et al., 2023). These tensions exist because forest management practices continue to support a strong export industry while growing concerns about biodiversity loss and sustainability raise questions about whether the current objectives are sufficient (Skogsstyrelsen, 2022). Presently, only a small portion of Swedish forest has been reserved to safeguard plant and animal species that are at risk of extinction against exploitation (Östlund & Zackrisson, 2000). Forest policy has also generally prioritised wood production meaning that economic aspects of sustainability and the interests of traditional forest sector actors continue to dominate (Beland Lindahl et al., 2017). In recent years, however, new pressures have however begun to reshape this landscape. Swedish forestry has been increasingly influenced by the urgent need to meet national and international climate goals and support the transition to a bioeconomy (KSLA, 2024). Certification schemes such as that of the Forest Stewardship Council (FSC)-certification have been created as a response to these environmental challenges, aimed at ensuring that timber production respects biodiversity, indigenous rights, and social values (Heder Brandt et al., 2023). It is partly in response to these sustainability and climate goals that digital technologies and data currently play a prominent role in Swedish forestry.

4.4.4 Digitalisation and the Use of Digital Technologies

Digital technologies have become increasingly prominent in Swedish forestry (Ivarsson, 2023; Koukouvinou, 2025). Digital technologies such as remote sensing, LiDAR, and machine learning have become routine parts of forestry management practices, allowing for cost-effective interventions, monitoring environmental conditions, and greater accuracy in decision-making (Urzedo et al., 2023). Forest work has also become more technologically advanced such that forest employees need to continuously update their skills and qualifications (KSLA, 2024). More recently, with the rise of artificial intelligence in different sectors, there has been a push towards smart forestry (Wang et al., 2024) for various activities such as assessing the health status of trees and simulating forest environments (Holzinger et al., 2024).

A key application of these technologies is in forest management planning, where scenario analyses are used to model and assess different forest management practices and their potential economic, societal, and

[11] https://omni.se/partier-oense-om-skogskrav-bruka-mer-inte-mindre

ecological impacts (Ulvdal et al., 2025). This analysis relies on data about the current conditions of forests to forecast how best to manage biodiversity and ecosystems over time (Ulvdal et al., 2025). Given that Swedish forestry must balance the growing global demand for wood with the multiple functions and uses of forests (Eggers et al., 2019), data have become an integral part of forestry decision-making and operations.

According to Gabrys (2020), forests provide vast amounts of data that can be captured through sensors which allow for real-time insights into forest dynamics. The data produced has also opened up new possibilities for a wider range of actors such as environmental activists and civil society to participate in conversations about the state of forests and challenge claims about the sustainability of Swedish forestry (Alarcon, 2023). Regardless, data and the digitalisation of forestry are not neutral processes. The ways in which environmental data are produced, shared, and used are far from straightforward and can therefore become sites of contestation on forest and land use (Gabrys, 2016). Thus, although digital technologies are often portrayed as unproblematic tools for addressing environmental challenges, they can facilitate data practices by experts that are exclusionary and deepen existing social and political tensions between different forestry interests (Gabrys, 2020). This calls for a rich empirical investigation of how digital technologies and data are used in organisational settings in forestry. The next sections explain and discuss how I collected and analysed data for such an empirical investigation situated in the context of Swedish forestry.

4.5 Data Collection

In this section, I present the case contexts for the different studies that resulted in Papers 1 to 3 and outline the data collection methods used in these studies. Paper 4 was a conceptual paper, meaning data were not collected. According to Myers (2020), it is important to select a case topic where the researcher has the potential to gain access and obtain relevant data. During my PhD, I was partly funded by Mistra Digital Forest (MDF)[12], a research programme focused on developing digital solutions to support sustainable forestry in Sweden. This research programme collaborates with several forestry companies across the industry, and this affiliation significantly facilitated access. Being funded by MDF played a crucial role in laying the groundwork for me to establish contacts and engage potential research participants for Papers 1 to 3. I was also able to gain access to respondents through a co-author on Paper 3 who acted as

[12] Mistradigitalforest.se

an insider researcher (Greene, 2014). For paper 4, the illustrative context was informed by the first three authors' prior experience conducting of research in the forest industry, which served as a basis for explaining and demonstrating the proposed design theory.

As my PhD journey progressed and evolved, so too did my interest and research focus. I began in 2021 with a broad interest in digital transformation, seeking to understand how companies in what I initially termed a traditional industry could leverage digital technologies and realise the value of digital transformation. However, given the conceptual ambiguity surrounding digital transformation, and debates in research and practice about whether it constituted a substantive shift or merely a buzzword, I began to explore how the concept could be unpacked and better understood. Through engaging in academic and practitioner discourses, it became evident to me that data and their role in digitalisation continuously emerged as a central theme of interest. Therefore, I refocused my research on the more granular aspects, specifically the role of data in creating digital representations for digitalising natural environments such as forests. Thus, the evolution of this focus is reflected in the cases that comprise the individual papers of this thesis.

In what follows, I provide further details on the case contexts and how and what data collected were collected for the empirical studies.

4.5.1. Study 1

Over the past decade, and even earlier, digital transformation has gained prominence in both the IS field and in practice as a means for digital technologies to disrupt established business models and radically transform businesses, to remain competitive in an increasingly digital landscape (Nambisan et al., 2019). The forestry sector is no different in trying to harness digital technologies for significant business improvements. However, the industry has largely been perceived as conservative and slow to adopt new technological advancements. Despite the general optimism among industry actors about the potential of digital technologies, existing studies highlight the persistence of traditional culture that influences both strategic thinking and day-to-day organisational practices. As a result, there remains a notable gap between the awareness of digital opportunities and organisational capabilities. Although the forest industry may not be at the forefront of innovation, recent years have shown a gradual but discernible momentum towards

adopting digital technologies, albeit not without multiple hiccups and challenges.

This study examined the role of senior management in multiple forestry companies in navigating the complexities of digital transformation within a traditional industry context characterised by scepticism and uneasiness towards change. In this study, interviews were conducted with nine (9) senior managers from five of the largest forestry firms in Sweden. The participants were sampled based on their expertise and involvement in their respective firms' digital transformation journeys. Their roles included Chief Digital Officer, Marketing Director, Innovation Manager, and Product Developer, among others.

4.5.2. Study 2

This study focused on the digital transformation journey of a forestry firm, pseudonymised as Alpha. Alpha is an international forest industry group organised into multiple business areas, including forestry, wood, and bioproducts. The company has a long history and outstanding reputation of exploring solutions to support forest planning and management, and industrial production processes in sawn timber and pulp production. Driven by its ambition of becoming increasingly sustainable and cost-efficient whilst meeting the needs of customers, Alpha has actively engaged in digital and innovative initiatives. In recent years, the company has also edged towards data-driven decisions to optimise its value chain and improve operational efficiency. In 2018, inspired by the buzz around digital transformation and the creation of a Chief Digital Officer (CDO) position at the group level, Alpha initiated several digital innovation efforts across its business areas. In the forestry division, henceforth Alpha Forest, a key focus on developing outward-facing digital services for forest owners emerged. Members of Alpha Forest, many of whom are small forest owners, hold a broad range of goals and ambitions for managing their forest estates, often to generate not only financial value, but also cultural, social, and ecological value for future generations. Many forest owners rely on forest management plans for responsible forestry and to preserve these values. Our investigation therefore focused on Alpha Forest's efforts to improve existing services and create new ones. Specifically, we investigated their efforts to digitalise their advisory services and improve how forest owners can manage their forests through an app.

Data collection for this study occurred in two phases. In the first phase, twenty semi-structured interviews were conducted. The interviews were

recorded and fully transcribed. They focused on understanding digital transformation and the role of digital technologies in operations. The respondents, who held positions at various levels in the organisation, were sampled as they were key stakeholders in Alpha Forest's digital innovation efforts. The respondents included business developers, area and regional managers, product owners, the head of products and services, the head of finance and IT, the CDO, an IT architect, and the head of data science and analytics. An initial analysis was conducted, and the preliminary findings were presented during the first two authors' two-day on-site visit to the company's Swedish headquarters. The second author in the paper also had a separate one day on-site visit. These on-site visits comprised formal and informal meetings with various individuals. Archival data in the form of internal confidential presentation materials and project documentation were collected in the first phase. In the second phase, and based on our initial analysis, three follow-up interviews were conducted. Publicly available information such as press releases were collected to supplement the dataset.

Table 2. *Data overview for Study 2*

Type of Data	No.
Recorded Interviews	23
Meetings with Alpha Forest representatives	4
Press releases	29
Other data sources	On-site visits, internal documents, corporate presentations

4.5.3 Study 3

This study, conducted as a multiple case study, investigated how data spaces were established and sustained to support innovation initiatives across three data-driven innovation projects in the forest industry. In recent years, the sector has faced increasing pressure to enhance environmental responsibility and ensure the long-term sustainability of forest management. This includes implementing measures to preserve biodiversity, strengthen the ecological role of forests, and maintain productivity. In response, the forestry sector and its adjacent industries in Sweden have intensified efforts to develop digital solutions aimed at addressing these complex issues. Developing these solutions necessitates cross-industry collaboration and frameworks for sharing data across organisational boundaries. These projects were selected due their use of data to develop digital solutions.

The first project focused on supporting forest owners in gaining a better overview of their forests and improving pre-commercial thinning, a critical silvicultural practice for sustaining forest vitality and long-term value. Although large forestry firms typically use advanced digital tools and data to identify thinning needs, small forest owners continue to rely on manual methods. These methods are becoming impractical for owners due to geographical distances. As a result, contractors hired for thinning must navigate forest stands without precision, leading to inefficiencies in forest management. Through a collaboration initiative involving diverse industry actors, the project aimed to develop a predictive model using data from remote sensing technologies such as LiDAR, satellite imagery, and open data. The model was designed to identify areas requiring thinning and forest management. Although an initial model was developed and demonstrated potential, it lacked robustness due to limited data inputs. For the model to be scalable, it required access to more extensive and diverse datasets, which in turn relied on sustained collaboration and data sharing among participating actors.

The second project focused on the forestry value chain and aimed to predict the properties of raw materials such as logs and sawmill chips to support informed decision-making. A key challenge encountered in the forest industry is optimising operations at sawmills using existing technologies such x-ray scanning machines. Despite the potential to maximise value, by-products such as sawmill chips have historically been subject to a lack of investment in quality assessment due to inadequate control in woodyards. The project, which was motivated by sustainability and efficiency goals, sought to link raw material characteristics to final products using a statistical model. This required extensive data collection from the various participating industry actors collaborating in the project. While the model was successfully developed, broader implementation across the industry was constrained by the lack of continuous industry-wide data sharing. Such a scaling approach would require collaborative infrastructure and coordinated data sharing among interested stakeholders and forestry firm.

The third project built on the results of the second project and extended the focus to the pulp industry, where consistency and quality are paramount. Pulp production requires forestry firms to supply raw materials with specific properties. However, the pulp sector faces persistent challenges in linking forest raw materials to final products and streamlining the digital declaration of raw materials properties. The project addressed this by modelling how variations in raw material properties influence production processes and quality. Industry actors

collaborated to share and integrate data from forestry firms, to be tested at pulp mills. Although the developed model was successfully validated in lab-based settings, real-time and seamless data and communication over a robust data infrastructure were required for real-world applicability.

For this study, data were collected from each project to assess the similarities and differences in creating and maintaining data spaces. Data were collected in two phases and access to the projects was facilitated by the fourth author's employment at the research institute that led the projects. The fourth author, who served as a project manager in Projects 2 and 3, acted as an insider researcher and provided valuable in-depth descriptions of the projects during recorded meetings. Also, access to Project 1 was gained through the fourth author's colleague, who acted as a project manager. Interviews served as the primary method of data collection and participants were purposefully sampled based on their direct involvement in and knowledge of the respective projects. In the first phase, ten interviews were conducted, recorded, and transcribed. These initially focused on the advantages and challenges encountered in data-driven innovation projects. Following a preliminary round of analysis, a second phase of data collection comprising twelve interviews was undertaken. This second round of interviews concentrated on understanding the role of data in the projects and how data were shared. The overall recorded interviews were supplemented with secondary data consisting of project reports, project funding applications, and media articles.

Table 3. *Data overview for Study 3*

Type of Data	No.
Recorded Interviews	22
Recorded meetings	4
Other data sources	Project funding applications, project reports, media articles

4.5 Data Analysis

In this section, I describe the role of data analysis in my appended papers and how the analysis approaches emerged and developed in each case. Data analysis involves the procedures applied to data to generate findings (Sarker et al., 2018). However, the process in thesis was far from straightforward and involved many iterative cycles between data collection, analysis, and theory. My process of data analysis in both my

individual papers and for the overall thesis has been a journey of self-reflection that led me to a central idea that resonated with my research interests.

As previously mentioned, I started my PhD journey with an interest in digital transformation which aligned with a growing discourse in IS research at the time (see Figure 1). Studying digital transformation in the forest industry pushed me to try to understand and theorise as to what about the research setting was so different. Here, I focused more on the context and used 'traditional industries' to distinguish between industries like forestry and born-digital companies such as Google. At this early stage of my PhD journey, the analysis was predominantly an exploratory process. However, I felt the term traditional industries was insufficient and did not fully capture what the digitalisation process looked like in forestry. Additionally, my focus shifted from the context alone to the broader phenomenon of natural environments. As I continued working on my individual papers, I began to question why digitalising natural environments was so different but had not pinpointed the appropriate concepts to answer these questions.

Then, in 2024, a paper by Goebeler et al. proved pivotal to my research process. Their focus on the role of physicality in digital innovation was enlightening and allowed me to conceptualise the aspects of natural environments that made their digitalisation distinct from that of other phenomena. Yet their paper also led me to challenge existing research and question how the characteristics of natural objects can also shape digitalisation processes. Drawing on Gregor's (2006) classification on the nature of theory, I describe my thinking as centred on understanding *how* and *why* some phenomena occur, with the aim of generating new insights. As I have continuously highlighted in this thesis, digitalising natural environments is a complex and pluralistic endeavour that involves multiple stakeholders. It therefore became important for me to examine how some key voices had been sidelined historically and how this problem could be addressed. In doing so, I moved beyond explaining why the digitalisation of natural environments unfolds in the way it does, towards offering normative prescriptions on how it should unfold to promote inclusivity. In Figure 1, I illustrate how my thinking and analysis evolved over time, showing the messy nature of the process as I progressed from focusing predominantly on the context to theorising about a broader phenomenon across my papers and this thesis.

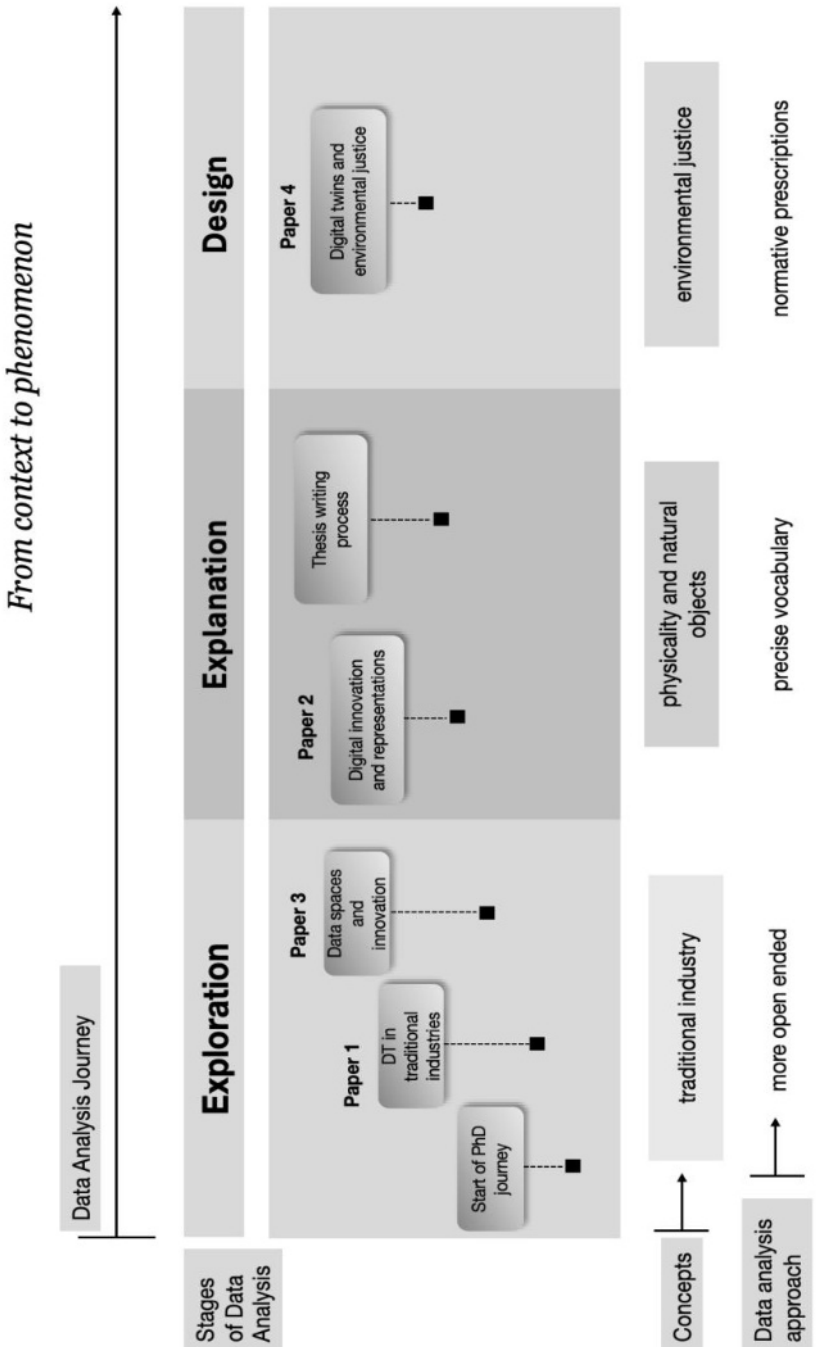


Figure 1.*Visualisation of the messy and evolving process of my data analysis*

Positioning my research within the interpretive paradigm required an openness to data and a willingness to revise preliminary assumptions, as suggested by Walsham (1995). Interpretive research often requires researchers to confront and reflect upon possible preconceptions they may hold about the phenomena under study (Klein & Myers, 1999). As with many researchers, my understanding of the context deepened over time. It often becomes difficult as a researcher to approach analysis with a blank slate. As I became more engaged in the Swedish forestry sector, my familiarity with the research context grew, which inevitably led to my forming of some theoretical preconceptions in initial stages of the cases.

However, this was mitigated by iterating between data collection and analysis to remain open to alternative interpretations and new insights that could emerge (Walsham, 1995). For example, in Paper 3, which focused on data spaces, our initial interest centred on why innovation projects failed to scale or be implemented in organisations. Yet, as the initial round of coding progressed, it became evident that data played an instrumental part in shaping those outcomes, and the focus gradually shifted in that direction. Although theory can be a useful initial guide during the iterative process of collecting and analysis data, there is also a risk of being stifled by it, if it is applied too rigidly (Walsham, 2006; 1995). In our case, we engaged with existing literature as a sensitising lens which loosely informed the second round of data collection. Still, we allowed concepts to emerge inductively from the data and be empirically grounded, rather than fitting them into pre-existing theoretical boundaries. This approach reflects Walsham's (1995) view that it is possible to engage with theory without becoming confined by it.

Although the appended papers differed slightly in how analysis was conducted, the overall approach was guided by thematic analysis. Thematic analysis was appropriate given its flexibility in accommodating various research questions, data collection, and approaches to meaning making (Clarke & Braun, 2017, p.297). In line with Braun's and Clarke's (2006) six-step guide, the research process for Papers 1 through 3, which employed qualitative methods, broadly followed these steps:

1) Familiarising yourself with your data.
2) Generating initial codes.
3) Searching for themes.
4) Reviewing themes.
5) Defining and naming themes.
6) Producing the report.

Data analysis began with familiarising myself with the data. Braun and Clarke (2006) suggest that researchers get immersed in the data during transcription and through repeated reading. As noted in Section 4.4, all interviews were transcribed after data collection and then carefully reviewed to familiarise myself with the content. For Paper 1, I used the qualitative analysis software Dedoose to enable collaborative coding and analysis by my co-author and me. Subsequently, for Papers 2 and 3, the qualitative data analysis software Atlas.ti was used for coding and analysis.

In Paper 1, familiarising myself with the data also meant becoming acquainted with the Swedish forestry context and industry-specific terminology used by respondents. This initial process gave me a broad overview of the state of Swedish forestry and its relationship to digitalisation. As I progressed through the subsequent papers (Papers 2 and 3), I became embedded in and increasingly conversant with the context. Although some cases in the individual papers were technically complex, the prior knowledge accumulated from Paper 1 helped me understand the inner workings of the industry. For example, concepts such as silviculture practices and precision forestry were still relatively new to me in Paper 3, but the grounding meant I could understand why forestry firms engaged in these practices and what they sought to achieve through their digitalisation. As a result, familiarising myself with the data became quicker and my understanding shifted from exploratory to a more grounded and specific engagement with the individual projects. This progression reflects my development as a researcher, and although the ideal aim was to remain untainted by assumptions about the context, I acknowledge that it was not possible to approach subsequent papers with a completely blank state.

In the next phases of data analyses, I began producing and organising my data into meaningful codes (Braun & Clarke, 2006). Coding is not simply a method to reduce data, but involves creating labels for important features relevant to addressing a research objective (Clarke & Braun, 2013). The coding process in my papers was iterative, relying predominantly on descriptive codes to capture aspects of the data key to answering my research questions. The initial and descriptive codes also provided insights and guidance into collecting additional data when needed. For example, in Paper 3, we engaged in open coding which then revealed that the role of data in the innovation projects was important to whether these projects succeeded or failed. As a result, we developed preliminary ideas about the codes and analysed newly collected qualitative data to understand how data were used and shared in the different projects.

Following the initial phase of generating codes, I searched for themes that revealed patterns in the data. This process varied across individual papers. For example, in Papers 1 and 2, the themes that emerged were either broadly informed or categorised using theory-driven concepts. In Paper 1, the themes forming the basis of our findings were informed by the framework of digital transformation described by Hanelt et al.'s (2021). Similarly, in Paper 2, we analytically engaged with Goebeler et al. (2024) and categorised our findings in accordance with their four roles of physicality.

Lastly, Braun and Clarke (2006) highlight that the end-point of research is to create a compelling and coherent narrative. My goal was therefore to communicate our research findings to various different audiences, including through journal articles and conference papers, all of which are appended to this thesis. I also presented working papers at workshops to get relevant feedback on my research. Given that my research setting is embedded in an industry context, it was also important to convey these findings to practitioners. To this end, I presented my research in poster sessions at an industry conference, which allowed me to engage in conversations with practitioners and gather their thoughts on my research. Below are the titles of these posters and workshop contributions:

1. Torto, M., Nylén, D., Nordström, M., Cartling Wallén, M. & Södergren, F. (2022, November). *Digitizing the Forest Sector: Insights from a Multiple Case Study* [Poster presentation]. Mistra Digital Conference 2022, Stockholm, Sweden.

2. Torto, M. & Kristofersson, A. (2022, May). *Sowing the Right Seeds & Harvesting Digital Transformation: A case study of the drivers and barriers to digital transformation in the forestry industry* [Poster presentation]. Mistra Digital Conference 2022, Umeå, Sweden

3. Torto, M., Nylén, D., Sandberg, J. & Kristofersson, A. (2022). *Enabling Structures for Digital Transformation: A Case Study of A Forestry Firm*. 4th Nordic Workshop on Digital Foundations of Business, Operations, Strategy and Innovation, Umeå, Sweden.

4.6 Ethical Considerations

As I employed qualitative methods like interviews in my research, it was important for me to consider a number of things. The research subjects to be interviewed must give their consent before the researcher can proceed

with the study. Prior to the start of an interview, in accordance with the Ethical Review Act (2008), the research subject must be informed about the overall plan behind the research, its purpose and the methods to be used, and the consequences and risks that the research project might entail, among other things. Informed consent was essential in my process. Because I investigated the different data used by organisations for innovation processes, it was key for me to provide research subjects with the assurance and understanding of what the end results of the research study could entail and how the results might influence their experiences in their organisations. These assurances were important as research subjects often divulged sensitive and business-critical information about their operations. In the process of conducting interviews, it is important to ensure that participants feel comfortable in the environment being set. This means that the participants' well-being must be taken into account. As such, I preceded each interview with the information that they could withdraw at any time, as interviews were voluntary.

I also sought consent to record interviews for the purposes of transcription and analysis. Thus, data collection could only proceed when my research subjects had consented to it, and this was documented. How data are used and stored following data collection is important. Due to the nature of interviews and the potential disclosure of personal and sensitive information by research subjects, I aimed to ensure that privacy was assured and maintained. To protect the privacy of my research subjects, data were always anonymised in such a way that there were no connections between the persons and the data. I also replaced all names, locations, and other identifiable information. As interviews could contain some personal data, the way the data were processed and stored were in line with General Data Protection Regulation (GDPR).

In accordance with the Ethical Review Act (2008) Sections 3–4, which stipulate the parameters under which a research project warrants an ethical review, I assessed the potential risks and sensitivity of my research and determined that it did not fall within this scope. Research projects which concern humans and biological material from humans or are concerned with using a method that affects a research subject mentally or physically among others are subject to ethical vetting (Section 4[13]). My research deals with actors and organisations situated in the context of forestry. Although ethical vetting may not be needed, it was still important to consider good research practices. As described by Vetenskapsrådet (2017), although some research can be conducted without an ethical review, researchers must still observe common research codes and reflect

[13] See section 4 of the Ethical Review Act (2008)

on their projects. Thus, I considered the ethical codes of informed consent and research integrity such as openness, fairness, and accuracy.

Ethical issues must also be considered in the analysis phase of the research. As my research followed an interpretivist tradition, it was important to approach the data without biases. In qualitative research, the challenge of objectivity and subjectivity comes into play. The researcher is faced with the task of managing subjectivity to establish rigour in the research process and must be aware of and monitor their personal ideologies in the research process (Holliday, 2007). It is crucial that the analysis is devoid of biases and preconceptions. Hence, the analysis should be carried out in a careful manner to reflect reliability in the quality of the research produced (ALLEA, 2017). In my position as a researcher, it is important to produce research that is honest, transparent, and accurate. Furthermore, in the analysis process, I aimed to be reflexive on my position as a researcher and acknowledge my presence in the setting and interaction with the phenomenon I was investigating (Holliday, 2007). Mason (2002) highlights the importance of 'active reflexivity' where researchers are constantly introspective and consider the actions and the role they play in the process.

Throughout my research process, I have considered how my personal ideologies might influence how I approached my research. Although I generally positioned myself as neutral in my research, I was aware of my personal ethos and what I considered meaningful, and how this influenced what I researched. Therefore, my research served not solely to understand a phenomenon but also explored what new knowledge I could create that would be valuable to the IS discipline and my case context, forestry. Therefore, I also considered how my findings and research would be received by different audiences and the impact they could have.

5. Paper Summaries

In this chapter, I summarise the four appended papers in my dissertation. Table 4 provides an overview of the papers based on the area of concern, research question, research problem, theoretical framing, unit of analysis, and contributions.

Table 4. *Summary of appended papers*

	Paper 1	Paper 2	Paper 3	Paper 4
Area of Concern	Digital transformation	Digital innovation and representing natural objects	Cross-boundary data sharing	Design of digital twins for environmentally-just forest management
Research Question	What are the drivers and managerial challenges in implementing DT in traditional industries?	How does the role of physicality affect digital innovation processes focused on representing natural objects?	How do actors navigate the challenges of establishing data spaces for innovation?	How can we design digital twins to enable environmental justice?
Research Problem	Managing and enacting digital transformation in traditional and physically anchored contexts	Developing digital innovation aimed at representing natural objects	Managing tensions between openness and control in data-driven innovation	Navigating competing priorities of how natural resources should be used and managed
Theoretical Framing	Digital transformation	Physicality in digital innovation	Data spaces and data-driven innovation	Rawlsian procedural justice
Unit of Analysis	Industry-level transformation	Forest management practices	Data-driven R&D projects	Digital twin design processes
Contributions	Empirical insights on how senior managers can make sense of and orchestrate digital transformation	Identification and illustration of three distinct characteristics of natural objects in shaping digital innovation processes	Identification of capabilities for both engaging in a data space and innovating to balance tensions between openness and control	Design principles prescribing how digital twins can be designed to fair and inclusive

5.1 Paper 1

Torto, M. & Kristofersson, A. (2023). Digitally Transforming in a Traditional Industry: A Senior Management Perspective. *Proceedings of the 56th Hawaii International Conference on System Sciences (HICSS).*

This study explores the complexities of digital transformation within traditional industries, with a specific focus on the Swedish forestry sector. It examines both the challenges and the accompanying motivating factors for implementing digital transformation (DT) in organisations.

Digital technologies have been proselytised to have transformative power and allow for changes to value creation paths and business models. DT is a prevalent goal in many industries, and organisations are facing increasing pressures to make technology-driven changes to their business processes and value propositions. Although DT is often discussed in 'digital born' contexts, this research emphasises the importance of exploring the phenomenon in other contexts such as traditional industries. Traditional industries, defined in this study as those centred on non-digital artifacts, often lag in digital maturity due to entrenched practices and legacy systems optimised for physical operations. Therefore, enacting change, which requires orchestrating capabilities, and restructuring operational processes and structures, are challenges that managers in these industries face. In this study, we investigated what the drivers and managerial challenges of implementing DT in traditional industries were. An aim was also to explore the phenomenon of DT in a context characterised by its non-digital raw materials. Drawing on an in-depth qualitative case study with senior managers from five of Sweden's forestry firms, our findings identify tensions between structure and strategy, customers' digital maturity and organisations' DT goals as dialectical factors for DT.

We discuss our findings by underscoring the crucial role of senior managers in spearheading DT initiatives. By leveraging their expertise and strategic vision, senior managers can guide organisations in navigating the complexities of DT. We also highlight how fostering a culture of innovation and adapting organisational structures to accommodate digital technologies may help solve persisting challenges and influence the success of DT efforts. Overall, the study provides valuable insights into the dynamics of digital

transformation in traditional industries. It also offers practical guidance for senior managers dealing with DT so they can unlock new opportunities for growth and innovation in their organisations.

5.2 Paper 2

Torto, M., Nylén, D., & Sandberg, J. (2025) Digital Innovation and the Representation of Natural Objects: The Role of Physicality. *Proceedings of the Thirty-first Americas Conference on Information Systems*

This paper explores how digital innovation unfolds when centred on representing natural rather than artificial objects, an area of research we found significantly underexplored in the current literature. Digital innovation scholarship has traditionally focused on artificial elements such as devices or built environments. In IS research, a substantial gap remains in understanding how innovation processes engage with natural objects, which are inherently different in their attributes, functions, and agency. Addressing this gap is important considering the mounting global challenges related to sustainability, biodiversity, and natural resource management. Our study aimed at addressing this by shifting the focus to natural objects and how their physicality enables and constrains the digital innovation process. To investigate this, we conducted a qualitative case study of a large Swedish forestry firm, Alpha, which was actively developing digital tools to support sustainable forest management. Drawing on the conceptualisation of physicality as a context, subject, vessel, and nexus, we examined how Alpha sought to digitally represent the forest's physical characteristics through a mobile app and how this shaped the innovation trajectory.

The paper contributes to digital innovation research by demonstrating how the agency, pace of change, and multifunctionality of natural objects shape the design and implementation of digital tools. These characteristics introduce unique challenges and necessitate new design logics for digital representation and user interaction. Through our case, we illustrate how digital innovation must account for the limited controllability of natural objects by enabling responsive, data-driven decision-making. We also highlight how the slow, unpredictable evolution of natural systems, contrasted with the rapid and iterative cycles of digital development, prompted a reconfiguration of traditional tools like forest management plans into dynamic, continuously updated digital services. Furthermore, we show that the multifunctionality of natural environments requires digital representations that can accommodate a plurality of user values and purposes. Additionally, the paper proposes

digital twins of nature as a promising avenue for future innovation, positioning them as boundary objects that can support real-time decision-making, stakeholder engagement, and participatory management of natural resources. Overall, our study contributes to the growing body of work on the physical dimensions of digital innovation. By engaging deeply with the physicality of nature, we show that innovating and representing natural objects require a fundamentally different approach than artificial and digital objects.

5.3 Paper 3

Torto, M., Nylén, D., Sandberg, J., Nordström, M., Cartling Wallén, M. & Södergren, F. Establishing Data Spaces for Innovation: Investigating Challenges and Responses in the Forestry Industry. (In manuscript)

With data increasingly recognised as a strategic asset and a catalyst for innovation, organisations face pressures to unlock their value, not only internally but also through external collaborations. This paper therefore explores cross-boundary data sharing for innovation, focusing on the Swedish forest sector. Enabling such collaborations for sharing data poses significant technical, organisational, and legal challenges. We investigate how actors navigate these complexities by forming data spaces – meta-organisational arrangements designed for the controlled and purposeful sharing of data across firms. Although the concept of data spaces has gained traction in recent literature, there is still a gap in empirical insights detailing how these spaces are established and governed in practice. To address this, we conducted a qualitative multiple case study of three data-driven innovation projects, gathering insights through interviews and from secondary data sources. These projects aimed at developing digital solutions in different areas along the forestry value chain.

Our findings revealed two sets of challenges that arise in forming data spaces: the intra-organisational and the inter-organisational. At an intra-organisational level, firms face issues related to data retrieval, quality, and access control. Although large volumes of data are often readily available, the data are frequently fragmented and difficult to extract or repurpose. Firms often lack critical infrastructure or expertise to efficiently retrieve relevant datasets. Additionally, issues such as incomplete datasets are common and can hinder progress. Access control also presents a challenge, as firms must ensure that appropriate safeguards are in place, particularly when dealing with personal or commercially sensitive data. At the inter-organisational level, challenges centre on building trust, ensuring regulatory compliance, and achieving data standardisation.

Firms participating in data spaces are often direct competitors, making it essential to carefully manage relationships and involve trusted intermediaries to facilitate data sharing while maintaining confidentiality. Legal constraints, particularly those related to antitrust regulations and the GDPR, also pose significant hurdle by restricting the types of data that can be lawfully shared. Furthermore, varying data structures and definitions across organisations make standardisation complex, although intermediaries play a key role in coordinating and synthesising data formats to enable compatibility.

This paper offers both theoretical and practical contributions. We identify three key capabilities that organisations must develop to engage in data spaces: (1) the ability to retrieve and manage data efficiently, (2) ensuring data quality and usability, and (3) governing access while balancing openness and control. The study also underscores the pivotal role of intermediaries in orchestrating collaboration, fostering trust, and navigating legal and technical requirements. By providing a detailed analysis of these processes, this paper broadens the understanding of data spaces beyond technical systems, showing how they involve intricate social and organisational arrangements that can support and sustain data-driven innovation.

5.4 Paper 4

Torto, M., Lindberg, A, Nylén, D., & Sandberg, J. Enabling Environmental Justice through Digital Twins: A Design Theory. (In manuscript)

This paper addresses the intersection of digital technologies and environmental justice by proposing a novel design theory for digital twins of natural systems. Digital twins – real-time, high fidelity digital models of physical entities – have been widely used in fields such as manufacturing, healthcare, and urban planning. As their application extends to natural resource management of systems such as forests, concerns have emerged about their potential to replicate and reinforce injustices by prioritising the interests of dominant stakeholders and marginalising those of others. The study addresses the structural injustices embedded in the digital infrastructures that shape environmental decision-making. These injustices manifest in the exclusion of minority groups from key decision processes, restricted access to relevant environmental data, and systems that are often non-transparent and inaccessible for non-experts to navigate. While existing research in information systems and environmental sustainability has explored the technical capabilities of digital twins and their potential for enhancing decision-making, less attention has been paid to their social

and ethical implications – particularly how they might be designed to include marginalised perspectives and support procedural justice. This paper therefore seeks to challenge this by advancing theory that enables digital twins to be designed as tools for inclusive, participatory planning and environmentally just outcomes.

This study, situated in design science research, draws on Rawlsian procedural justice theory which emphasises fairness, equity, and the protection of basic liberties. Rawls' principles, particularly the focus on arrangements that benefit those least advantaged, provide a lens for conceptualising how digital systems can be designed not solely for efficiency but also to promote just outcomes. To operationalise such a system, our paper proposes a set of modifications to the conventional digital twin model, to align with Rawlsian justice principles. These adjustments address critical elements such as data collection, modelling and visualisation, user accessibility, and decision-making processes. Together, they ensure that all stakeholders, and not just those with technical expertise or institutional power, can contribute to understanding and influencing the planning of natural resources. We translate these modifications into a set of design principles, which include enabling diverse data representations, supporting pluralistic forms of modelling, facilitating discourse among stakeholders, and fostering shared visions. We illustrate our argument in the context of Swedish forestry, a rich empirical backdrop marred with competing stakeholder interests, ranging from economic productivity and biodiversity conservation to indigenous cultural practices and recreational use.

This paper argues that achieving environmental justice is not a one-off outcome but an ongoing, iterative process. Digital twins must therefore be dynamic and adaptable, continuously updated to reflect evolving stakeholder perspectives, environmental conditions, and societal values. We offer contributions to both IS research and the broader discourse on digital sustainability. The proposed design theory offers conceptual clarity and practical guidance when developing digital tools that promote fairness, inclusivity, and long-term sustainability.

6. Discussion

Digital technologies are no longer passive resources used for internal operations but a source of strategic and transformational change across organisations and industries (Alaimo & Kallinikos, 2024). As digital technologies are introduced and embedded into contexts typified by tangible products and physically anchored phenomena, actors begin exploring how they might leverage these digital technologies in meaningful ways. A prominent approach has been to create digital artifacts and representations of physical phenomena (Gradillas & Thomas, 2025; Østerlie & Monteiro, 2020). IS research has examined digital representations of phenomena across a range of empirical domains such as contemporary work practices (Jonsson et al., 2019), environmental monitoring (Monteiro & Parmiggiani, 2019; Østerlie & Monteiro, 2020) and more recently experiential customer offerings (Lorenz et al., 2024). These studies are situated in varied contexts such as iron ore mining, marine environments, and wine tasting yet they share an important characteristic: they cannot be fully digitised due to the inherent physicality of the phenomena and environments in which they occur.

While this body of work demonstrates how digital technologies enable the digitising of physical phenomena, it has paid comparatively less attention to theorising or empirically examining how the physical attributes of such phenomena influence or shape their digital representations. Research on environmental monitoring in marine environments (Monteiro & Parmiggiani, 2019; Østerlie & Monteiro, 2020) begins to address this issue, yet the broader IS discourse continues to focus predominantly on technical artifacts and designed objects. As such, there remains a limited understanding of how natural environments, and their distinct physical characteristics, become digitally represented as part of digitalisation efforts. This thesis addresses this gap by offering empirical insights into how efforts to digitalise natural environments at the organisational, industry, and societal levels extend beyond merely introducing new technologies or developing digital products and services. In this thesis, I have sought to answer the following research question:

How do sociotechnical interactions shape the digitalisation of natural environments?

Drawing on an in-depth investigation of Swedish forestry, I highlight that digital representations of natural environments are shaped not only by

organisational objectives or user needs but also by the physical attributes of the natural objects being represented and digitalised. In this way, the physicality of natural environments conditions and shapes the digitalisation process. In this penultimate chapter, I discuss and illustrate how the empirical findings from the appended papers illuminate the dynamic process of digitalising natural environments. I begin by discussing how the digitalisation of natural environments entails creating digital representations of natural objects to serve different purposes at various levels of analysis. I then unpack three different perspectives on data and how these views influence the trajectory of digitalisation. Lastly, I detail the limits and potential shortcomings of digitalisation.

6.1 Natural Objects and Digital Representations

The digitalisation of natural environments triggers changes and encounters challenges at the industry, organisational, and societal levels. At each of these levels, digital representations of natural objects serve different purposes, from improving value chain operations to supporting customer-facing initiatives and broader public engagement. In what follows, I will discuss these levels, each in turn.

First, the digitalisation of Swedish forestry has involved industry-level initiatives focused on transforming the value chain. As a starting point, actors across the industry have deployed digital technologies such as sensors, drones, and remote sensing technologies to digitise pockets of value chain activities, including forest planning, harvesting operations, and adjacent mill production. Rather than focusing on a single firm, these initiatives seek to integrate data through collaborations between diverse industry actors. One such initiative was studied in Paper 3. One of the projects studied in Paper 3 (Project 1), focused on digitising *forest planning*, an activity where both large forestry firms and small forest owners prepare specific forest stands for harvesting. The project constellation knew that digitising these activities was particularly challenging for small forest owners. Large forestry firms, for instance, use drones and have access to other advanced technologies to carry out their forest planning goals. In contrast, small forest owners typically do not possess such resources, and in addition often reside far away from their forest. For the small forest owners, collecting data for silviculture practices and forest planning has therefore been a labour-intensive and time-consuming task. It has involved manually navigating uneven terrains to locate forest stands in need of pre-commercial thinning. In this way, the spatiotemporal characteristics of the forest environments,

including the geographical location and spatial dispersion created significant challenges for collecting data that would serve as a basis for creating digital representations of forest conditions. It was this problem that motivated the formulation and execution of the project.

The challenges that the actors studied in Paper 3 faced in terms of collecting data triggered the creation of digital solutions across the value chain, capable of representing forests more effectively. This development has been highlighted by Gibson and Walford (2025) in forestry research focused on digitalisation and is also a point that Goebeler et al. (2024) make in IS research about the role of physicality. In this way, the characteristics of natural objects can either drive or hinder digital innovation initiatives as part of digitalisation goals. Given that natural objects (forests) possess various physical attributes such as spatiality, location, and mass (Ghazali, 2006), capturing these attributes required detailed patterns of tree density and tree height to provide a faithful spatial profile of the forest environment. Such spatial profiles required the use of high-cost and advanced technologies such as LiDAR and aerial satellite imagery to collect data and develop detailed maps for representing vegetation and forest terrains, thereby fulfilling the main goals of Project 1. My findings also show that digitally representing forests can be particularly challenging because forests are dynamic natural environments. Forests evolve over time and are influenced by external factors such as climate change (Andersen & Wicken, 2021). As Moreno & Etxeberria (2005) suggest, natural environments possess agency shaped by evolutionary processes. Due to this, factors such as weather variations, cloud cover in satellite imagery, and geographical differences affecting tree growth over time could not always be represented digitally in Paper 3. Instead, these conditions were accounted for through the data points used to construct map representations of forest stands.

Second, the digitalisation of Swedish forestry has unfolded at both organisational and project levels. Rising customer expectations and competitive demands have intensified pressures for organisations to digitalise their operations (Guggenberger et al., 2026) and the Swedish forestry sector is no exception. Still, rather than concentrating mainly on internal organisational processes, many initiatives have focused more on identifying opportunities for new products and services. As Hinings et al. (2018) note, digital innovation increasingly reshapes the nature of value creation by enabling organisations to orchestrate new products, services, and business models within their contexts. These dynamics are especially evident in Paper 2, where rather than focusing on the entire value chain

as in Paper 3, forestry firm Alpha directed its digital transformation efforts towards customer-facing initiatives. In response to changing expectations, Alpha began to explore new pathways for value creation, recognising that customers had become accustomed to digital products and services in other industries and in everyday life. To better understand their customers' emerging needs, the company initiated a data collection effort through an online platform to gather customer insights about challenges and preferences on forest management activities. This approach aligned with existing research suggesting that organisations can identify directions for innovation by engaging with user narratives, which often signal evolving customer expectations and emerging needs (Nambisan et al., 2017; Hund et al., 2021).

Building on these insights, Alpha's digital innovation initiatives were therefore centred on creating digital representations of forests. IS research has long posited that objects which occur in the physical or social world can be mirrored through digital representations (see Lyytinen, 2022). However, Turnbull et al. (2023) challenge this assumption by arguing that not all aspects of physical phenomena can be neatly translated into digital representations. In the process of creating representations, elements of the physical world are inevitably simplified or omitted, creating reduced versions of the phenomena. This was evident in Paper 2, where the characteristics of natural objects such as their agency shaped what could be captured and represented digitally, influencing the scope and direction of Alpha's customer-facing initiatives. As Burton-Jones and Grange (2013) note, digital representations can only capture aspects of the physical referent as *faithfully as possible*, and the findings in Paper 2 illustrate how the attributes of natural objects set limits on what can ultimately be represented.

Data also emerged as a critical component of Alpha's innovation process (Paper 2). As Yoo et al. (2024) note, data have become a central frontier in digital innovation (p.5). In Alpha's case, different aspects of the forest and its ecologies could be rendered into datasets which were used to establish patterns in the changing conditions of forests. By deploying digital technologies such as sensors, data on what Gibson and Walford (2025) describe as the 'materiality of nature' – including arboreal environments and forest structures – can be collected and analysed. Digital technologies also enable monitoring of the natural environment in areas that were previously inaccessible, support real-time sensing, and provide insights into how people interact with natural environments (Arts et al., 2015; Urzedo et al., 2023). Paper 3 illustrated how the dynamic and complex characteristics of natural environments present particular

challenges at the industry level, where the large forest areas and multiple actors involved make real-time monitoring at scale difficult. At the organisational level, however, where the scope is smaller and more contained, real-time monitoring is more feasible. These capabilities were reflected in Paper 2, where Alpha leveraged homogenised datasets describing the physicality of forests to develop new digital services and products (Kohli & Melville, 2019). A main element of this outcome was the development of a customer-facing product that digitised and supported forest management plans. It took shape in the form of an app which allowed forest owners to access information about their forests and participate more actively in decisions surrounding forest management. This suggests that through digital innovation, the boundaries between physical and digital materialities can converge, reshaping how customers engage with offerings and experiences (see Yoo et al., 2012; Hund et al., 2021). Paper 2 thus illustrated the claim of Turnbull et al. (2023), that digital technologies and representations enable new forms of engagement with nature for actors that were formerly inaccessible to them. Through my findings, I identified that the digitalisation observed in Paper 2 at the organisational level was not solely about developing new services to enhance organisations' competitive positions. It also transformed customers' interactions with their forests by mediating those interactions through digital representations (cf. Jonsson et al., 2018).

Third, digitalisation at the societal level is equally significant, especially in the context of natural environments where multiple stakeholders hold competing interests and values. In Paper 4, we sought to provide normative prescriptions regarding the digitalisation of natural environments with the societal level in mind. Although forestry is a key economic driver in Sweden and accounts for 9–12% of its GDP (Skogsindustrierna, 2025), the fact remains that Swedish forests hold protective, environmental, and cultural values for various groups. Because forests cut across economic, social, and political domains, the diverse uses attributed to them may not always be compatible (Grimble & Wellard, 1987). Nature – in this instance the forest – is therefore recognised and interpreted from different perspectives (Gabrys et al., 2025). At the societal level, discourses surrounding the use of natural environments often involve tensions between competing perspectives on their use and governance. Research suggests that digital representations shape how members of society perceive and engage with nature, while also creating opportunities for allowing under-represented voices to participate and intervene in natural resource use and management (Gabrys et al., 2022; Arts et al., 2015). Similarly, Luque-Ayala et al. (2024) highlight the potential of digital tools to enable societal actors to challenge

hegemonic conversations surrounding nature. Given the inherent unpredictability of natural environments highlighted in Paper 2, digital representations can be used to model and predict environmental dynamics (Westerlaken, 2024). A key advantage of digital representations lies in their ability to enable experimentation and intervention of natural environments without directly affecting the physical phenomena they represent (Bork et al., 2024). Despite their potential benefits, they have been subject to criticism. Scholars argue that tools and solutions for decision-making can be fragmented, economically oriented, and often designed *for* diverse actors rather than *with* them (see Westerlaken, 2024; Arts et al., 2015; Tzachor et al., 2022). These critiques resonate with the arguments in Paper 4, where we advocate for more inclusive approaches to creating digital representations to address the environmental injustices that arise when the perspectives of minority stakeholders are sidelined.

Digital representations have been proselytised as being capable of capturing biological systems and creating non-exclusionary outputs (Bruyneseels, 2020), yet in Paper 4 we argue against this as an overly optimistic view when applied to natural environments. Research has consistently shown that data used to construct digital representations are not neutral and do not paint a comprehensive picture of natural environments, as decisions about how nature should be represented and governed are deeply entangled with political interests (Nost & Goldstein, 2022). Paper 4 expands on this argument by demonstrating that the act of selecting which aspects of the forest are represented is not neutral but shaped by the dominant interests and inherent biases within Swedish forestry. Forests in Sweden are at the centre of ongoing debates where conflicting discourses regarding how they should be used and conserved are present. As a result, forests cannot be deemed apolitical. Westerlaken (2024) similarly highlights that the choices about which biodiversity indicators or data on natural objects are prioritised in digital representations may exclude other important forms of knowledge about natural environments. These dynamics raise important concerns regarding bias and exclusion in natural resource management. Biases and exclusionary practices embedded in data, and the resulting representations may deepen existing inequalities, raising questions about who ultimately benefits from digitalisation (Turnbull et al., 2023). To address these challenges, Paper 4 argues that the data used to construct digital representations of natural environments should incorporate more diverse and minority perspectives. For instance, representations of Swedish forests should incorporate aspects that are significant for indigenous communities, such as the Sámi population's use of vegetation

for reindeer herding. Additionally, we emphasise in Paper 4 the need for digital representations to reflect values that recreationists find important, such as mushroom and berry picking.

As digital representations increasingly inform environmental policies and decision-making (Westerlaken, 2024), it is essential to consider who participates in creating them. Ensuring that dominant actors are not the only participants is crucial for confronting and addressing environmental injustices that reproduce existing power dynamics (Gabrys et al., 2025). As acknowledged and discussed in Paper 4, the making of digital representations involves sociotechnical processes embedded within broader sociopolitical contexts and is therefore susceptible to societal pressures that influence how natural environment are represented. This observation aligns with the argument by Luque-Ayala et al. (2024) that digital technologies do not only observe physical phenomena but actively participate in constructing them. In this sense, decisions about what becomes actualised as a digital representation do not simply mirror the physical world but construct nuanced versions of natural environments. Setting out to articulate a design theory, we therefore questioned uncritical narratives of digitisation through arguments in Paper 4. Paper 4 reiterated claims in existing research that knowledge politics are manifested in how phenomena get represented. Fundamentally, the basis for the digital representations of natural objects is conditioned by the types of data that exist and for what purposes they are collected. Consequently, digitising physical phenomena becomes a contested and political practice, as data reflect the institutions and interests of dominant stakeholders (Tretter & Burns, 2023). The interests of such stakeholders in forestry are predominantly centred on commodification. This then brings forth the questions of whether digital representations are seen as enabling only new forms of commodification of natural objects (Luque-Ayala et al., 2024) and how do we move to a more pluralistic practice on data.

6.2 From Commodities to Relations in Natural Environments' Digitalisation

Data are central not only for constructing digital representations. The aspects of natural environments that get quantified and abstracted into data are essential for driving innovation and ultimately shaping the digitalisation trajectory of natural environments. Hence, the kinds of data that organisations prioritise, and the purposes for which data are

collected, will have implications on what is considered focal in digitalisation and how digitalisation unfolds in an industry.

In Chapter 3, I drew on Jarvenpaa's and Markus' (2020) three perspectives to illustrate how data are conceptualised, understood, and used in organisations. According to Jarvenpaa and Markus (2020), data can be viewed from a commodity, processual, or relational perspective. These perspectives shape the processes through which digitalisation is enacted. Extending beyond the technical process of digitisation, research has highlighted how digitalisation involves reorganising social and technical structures in pursuit of specific societal, organisational, or individual goals (Baiyere et al., 2023). In what follows, I discuss how the three perspectives on data manifested across my empirical cases and how they condition and shape the digitalisation of natural environments. Grounded in my empirical investigation, Table 5 provides an overview of these manifestations, organised into three salient categories: structures, actors, and processes.

Table 5. *How data views shape the digitalisation of natural environments*

	Commodity view	Processual view	Relational view
Structures	Rules and norms for determining which data about a phenomenon are collected and represented are orientated towards economic goals, competitive advantage, and developing new business models.	Rules and norms shape how phenomena are forged into organisational reality as data. Structures such as data spaces emerge to govern data production and reuse across organisational boundaries.	Rules and norms for tailoring data towards intended users are shaped by diverse stakeholder values, requiring pluralistic approaches to avoid reinforcing power imbalances or excluding minority interests.
Actors	Chief digital officers and their peers focus on operationalising data about natural environments as strategic resources to improve product	Production managers and logistics engineers, for example, selectively sense and capture aspects of natural environments based	Researchers and domain experts work to capture aspects of phenomena based on the goals and

	offerings and gain monetary value.	on their subjective knowledge and institutionalised perspectives on what is considered valuable.	interests of intended users, with various other actors recognised as key contributors in defining which data are relevant.
Processes	Data activities are directed toward securing desired goals that transform data into goods and services that generate value for individual firms.	Data are transformed, repurposed, and shared through negotiated activities involving anonymisation and standardisation.	Data activities are adapted to be participatory and inclusive to make data meaningful and actionable for diverse users and communities.

Given forestry's major role in Sweden's national economy, it was unsurprising that my findings indicated that digitalisation initiatives were predominantly oriented towards economic outcomes. This economic orientation was a representation of the commodity view and became evident in the data organisations prioritised for developing new goods and services. When data are viewed from a *commodity* perspective, they are understood as standardised and unambiguous (Alaimo et al., 2020), allowing them to be treated as objective and liquid assets for deriving monetary value (Xu et al., 2023). In this sense, data are positioned as strategic resources that support decisions and improve product offerings for increased competitiveness. In Sweden, competition is salient in the forestry sector as there is a small number of large forestry firms who offering similar products and services to customers such as small forest owners. Across my empirical material, maintaining competitiveness consistently emerged as a key driver of digitalisation and digital transformation in Swedish forestry. For example, in Paper 1, senior managers in forestry firms described remaining competitive and being profitable as central goals behind digital transformation. This observation aligns with extensive research on digital transformation which emphasises how organisations leverage digital technologies to gain competitive advantage (see Vial, 2019; Legner et al., 2017). To achieve this, managers in the forestry firms recognised that achieving digital transformation would require novel business models to identify new value streams as well as a shift towards data-driven decision-making.

The need for transformation was also evident in Paper 2. In that study, forestry firm Alpha acknowledged the importance of using data to identify opportunities for value creation. At Alpha, the Business Development department amassed and analysed large amounts of data to create digital solutions that would position Alpha as an attractive and contemporary service provider of forest management to customers and strengthen their competitive edge in the industry. In instances where the primary focus is on generating economic value for individual forestry firms, what counts as data and what is consequently collected about natural environments is shaped by the interests of organisational actors such as managers who operationalise these data for future profit. This finding adds nuance to Jones' (2019) argument that what becomes recognised and selected as data reflects what actors consider important. However, translating data into monetary value will require new roles, capabilities, and knowledge that may not always be readily available within organisations. Jones (2019) similarly highlights that organisations must make pragmatic decisions about which data are worth capturing and recording. Paper 2 shows how this need prompted organisational restructuring, including hiring a Chief Digital Officer (CDO) who was tasked with developing a digital strategy and identifying how existing data could support digital transformation across the firm. The introduction of a CDO by Alpha echoes findings from Horlacher's and Hess' (2016) study, which showed how CDOs proactively drive digital transformation by devising and implementing digital strategies. The role also facilitated cross-functional collaboration, enabling actors to jointly develop initiatives aimed at advancing the firm's digital transformation efforts.

Whereas Papers 1 and 2 highlighted firm-level competition and strategic goals, Paper 3 revealed a different dynamic. In the three projects I studied in Paper 3, actors at an industry level engaged in data sharing to improve operational efficiency and make more data-driven decisions in the value chain. This finding is in contrast to those of Gelhaar et al. (2021), who suggest that competitive tensions may discourage organisations from sharing data. In the Swedish forestry sector, firms were willing to collaborate and share data despite the competitive environment. Interestingly, the goal for sharing data was not anchored in short-term financial returns but in generating value over a long-term horizon. Such long-term planning is characteristic of industries centred on natural resource extraction and management, where prioritising short-term economic gains risks the long-term depletion of natural environments and ecosystems (Nelson & Scorah, 2021). Other research has shown how firms can transform data into assets and commercialise them (Parviven et al., 2020), whereas Paper 3 revealed that for the forestry firms

collaborating at the industry level, short term monetary gain was not the primary objective.

In the above, I have discussed how viewing data from a commodity perspective reflects the monetary goals of firms in the Swedish forestry industry. However, understanding why certain data become valuable for digitalisation is only one side of the picture. We also have to consider how such data even becomes useful for decision making and organisational goals in the first place. Thus, when data are viewed from a *processual* perspective, emphasis is placed on the ongoing process of how data are transformed, interpreted, and reused for organisational goals (Jarvenpaa & Markus, 2020). As Alaimo and Kallinikos (2024) note, 'defining data only as a ready-made resource says little concerning the modalities by which these resources are produced or repurposed, made sense of, and aggregated' (p. 62). What therefore becomes important is not only the outcome of data use but also the practices through which data are actualised into digital representations (Aaltonen & Tempini, 2014) and materialised in organisational settings. Rather than assuming that encoding physical phenomena into digital artifacts is a purely automated process, the processual view directs attention to the sociocultural and organisational activities and choices through which such phenomena are forged into organisational reality as data (Gradillas & Thomas, 2025; Aaltonen & Stelmaszak, 2024). In this sense, the very process through which phenomena are made into data institutionalises particular forms of knowledge about the world and becomes the basis for individual or collective action (Alaimo & Kallinikos, 2024).

My empirical findings illustrate how these processes unfold in the digitalisation of natural environments. In Paper 2, Alpha's customer-facing app was shaped by the contextual cues and their newly acquired knowledge about the small forest owners. In Sweden, forests have historically been passed down through generations, so the demographic of forest owners has been older. However, Alpha recognised that its customer base had shifted to younger and more urban forest owners who no longer lived near or were able to regularly maintain their forests. These characteristics informed how they selected and interpreted forest data to design the app's functionality and create digital representations. Yet the act of producing and framing such contextual data is not neutral; it is selective and socially shaped (Jones, 2019). By developing the app, Alpha effectively established boundaries on what contextual aspects would be quantified into data. As Dalton et al. (2020) argue, this boundary-making is a way through which actors set limits on what can be known and secured in pursuit of their strategic ambitions. From this point of view,

transforming data into useful organisational knowledge is socially embedded and, as Alaimo and Kallinikos (2024) describe, constitutes a by-product of human activities.

Further, the *processual* view of data extends beyond selecting and using data for organisational objectives; it also encompasses ensuring the accuracy and completeness of data to be aggregated and repurposed across different contexts. In their study on sand monitoring, Østerlie and Monteiro (2020) describe how oil and gas companies repurposed sensor data to create simulations of sand erosion as it was difficult (if not impossible) to monitor the wear and tear of equipment caused by erosion in real time. Their process hinged on the accuracy of erosion algorithms to prevent distorting the digital representations of the phenomenon. A similar challenge was faced in the forestry context where actors sought to forecast the properties of trees prior to processing into sawmill chips. In Paper 3 (Projects 2 and 3), addressing this issue required sharing, repurposing, and interpreting accurate and timely data from multiple sources across different forestry firms, which gave rise to data spaces.

Indeed, the *commodity* view is apparent across the projects studied in Paper 3, where the focus was predominantly on increasing efficiency through digital innovation. However, as the firms engaged in data sharing the *processual* and *relational* views became salient. Thus, although Jarvenpaa and Markus (2020) present the three views on data as distinct, my findings reveal that they are not mutually exclusive – in practice, they overlap. This is especially evident in terms of how the *processual* and *relational* views intersected in the empirical material of Paper 3. As sharing data was approached with understandable hesitance by some forestry firms, a neutral actor – defined in the literature as an intermediary (see Janssen & Singh, 2022; Schweihoff et al., 2024) – was tasked with facilitating data sharing and use. Although the processual view emphasises the socially embedded nature of data production and design (Aaltonen & Stelmazsk, 2024), my findings in Paper 3 reveal that these practices are not only socially embedded but also institutionally negotiated across organisational boundaries, particularly when data sharing occurs among competing firms. Paper 3 demonstrates the institutional friction that arises when data must cross organisational and competitive boundaries. The data that were eventually included in the data spaces to predict qualities of tree species did not emerge as a straightforward mapping of reality but through a series of negotiations mediated by the intermediary. For example, harvester data on timber logs which provided the basis for forestry firms' end-products could only be shared if they were anonymised and stripped of key data values. These

restrictions were necessary because the participating forestry firms were collaborating with direct competitors.

When data becomes recontextualised and reused in multiple contexts, Jones (2019) suggests that new questions about how these data should be governed inevitably arise, a phenomenon that manifested as the data spaces emerged. In Paper 3, the forestry firms raised critical concerns about data access and regulation, questions that had not typically arisen when data were strictly internal to individual firms. When data are reinterpreted and repurposed for new outcomes, concerns about how they should be regulated become more prominent. Consequently, data practices shift as the number of involved actors increase, especially when those actors are competitors. This shift necessitated the emergence of new forms of data governance among the forestry firms, resonating with prior suggesting research that, when the scope of data sharing becomes inter-organisational, robust mechanisms and structures for data access and protection will need to be established (Abraham et al., 2019; Jagals & Karger, 2021). This finding suggests that regulatory frameworks must evolve in tandem with the repurposing of data.

In the literature on data governance, researchers propose that formal structures for data sharing are essential because they codify accountabilities such as transparency, standardisation, privacy rights, and access (Jagals & Karger, 2021; Vial, 2023). Within the forestry sector, these structures for inclusion and exclusion emerged in the data spaces as a safeguarding measure to ensure data protection. Furthermore, because individual forestry firms may selectively sense and capture attributes of the forest based on their subjective view of what is relevant or valuable, standardising rules and making collective decisions became paramount. Such a finding corroborates the suggestion of Guggenberger et al. (2025) that decision-making within a data space should be grounded in a consensus approach among all the participants. In this way, the risks associated with potential biases or subjectivity in data can be mitigated. A notable finding from Paper 3 was that although the intermediary formalised rules and norms for data sharing, the functioning of the data space ultimately rested on a foundation of relational trust – which Poppo et al. (2016) describe as arising when actors develop shared values and mutual understandings. In Swedish forestry, this trust had been built up over time through informal relationships and past interactions among forestry firms. As a result, even though these firms were competitors, their shared understanding that digitalisation initiatives served the collective betterment of operations in Swedish forestry helped temper competitive frictions and enabled collaboration.

Unlike the *commodity* and *processual* perspectives, which focus on commercialisation and data production, respectively, the *relational* view emphasises the importance of tailoring data to the diverse needs and experiences of those who use them. Under this view, data may hold financial, cultural, or epistemic value, and these different values shape what data are assembled, shared, and how it is interpreted (Leonelli, 2020). In Section 6.1, I highlighted that Swedish forestry comprises diverse actors whose views on the forest are politically and ideologically conflicting. Tensions often emerge from trying to balance economic and ecological goals around forestry. Therefore, adopting a *relational* view towards natural environments, which holds value to different stakeholders, requires data that are collectively agreed upon by those stakeholders, as Alaimo et al. (2020) emphasise. I argue that a foundational step is identifying how to invite and get all key stakeholders involved into this dialogue. As illustrated in Paper 4, digital twins have the potential to facilitate this inclusive discourse. Here, the expansiveness of the data determines how accurately the forest is represented. The commodity perspective, as illustrated in Paper 1, treats data as objective and faithful representations of reality (Jarvenpaa & Markus, 2020), aligning with the notion of Vera et al. (2019) an 'extractive logic' in which data are framed as a raw material to be harvested within dominant corporate and government data practices. In Paper 4, we critically examine this logic and argue that a *commodity* view on data prioritises the economic interests of dominant stakeholders while systematically excluding minority interests. This resonates with Jones' (2019) caution that data does not mirror the world transparently and therefore must be approached critically rather than passively accepted at face value.

Throughout this thesis, I have emphasised that data are non-neutral and can embed systemic biases that reinforce injustices. A *relational* view is thus particularly important for digitalising natural environments, where systemic challenges run the risk of being entrenched and reproduced through the data collected and prioritised. This point aligns with the argument of Turnbull et al. (2023) that environmental data can play an active role in exclusionary processes that construct ecologies. As a response, in Paper 4 for instance, we suggest that the data parameters that are used to construct digital twins of natural environments should be broad and inclusive to dismantle injustices that might otherwise be perpetuated. This finding extends Leonardi's and Treem's (2020) work by asserting that transparency in who has a voice in defining what data are considered useful is essential to achieving equity and a fair representation.

Ultimately, moving towards a *relational* perspective on data for the digitalisation of natural environments, requires a reckoning on what current dominant datasets recognise, legitimise, and reproduce. In Paper 4, we argue that existing forestry datasets are shaped by normative frames that sustain an idealised and ostensibly objective view of forests. This supposed objectivity is solidified through the existing data practices and rules of data quality, which – as noted by Aaltonen and Stelmaszak (2024) – highlight both the successes and inherent failures of representational work. In this configuration, power imbalances and injustices only become reified, creating a cycle of systematic exclusion. To break this cycle, a pluralistic approach to forests and broader natural environments is required. As Gabrys et al. (2022) suggest, this must involve acknowledging indigenous and traditional communities as key contributors in cultivating and envisioning the future of our forests, finally centring the voices that have historically been sidelined in data and representational work. In this thesis, I argue that a monolithic perspective on the digitalisation of natural environments fails to reflect and capture their inherent diversity. In line with Walker et al. (2018), I contend that the onus is on organisational actors to rethink data practices to enhance visibility and accessibility in ways that account for the diverse values that are attributed natural environments.

6.3 On the Limits of Digitalisation

In this thesis, I have explored how the digitalisation of natural environments unfolds. Digital technologies increasingly play a central role in reshaping values and markets surrounding natural environments and are routinely enrolled in sociotechnical practices to monitor and transform (Gabrys et al., 2025). The grassroots of digitalisation often begin with converting physical phenomena into digital artifacts (Tilson et al., 2010; Baiyere et al., 2023; Gradillas & Thomas, 2025) and my research demonstrates that even this initial act is not simply technical but is also a socially embedded process shaped by assumptions and the vested interests of organisational actors.

In forestry, where the physicality of the natural object cannot be abstracted away, digitalisation is grounded in attempts to represent the physical world once data have been captured (Aaltonen & Stelmaszak, 2023). Yet, as discussed in Section 6.1 and particularly illustrated in Paper 2, the vitality and complexity of forests cannot be fully modelled due to the dynamic and interdependent ecosystems characterising forest environments (von Detten, 2011). Similarly, Leonardi and Treem (2020),

drawing on glaciers as an example, show that although natural environments can be measured and monitored, processes such as measurements construct objects rather than capturing them in their entirety. Digital representations are therefore never clones of the physical world but rather a product of strategic choices by decision-makers about what to represent and what to omit. At the organisational (micro-) level, such simplification is often purposeful and a necessary survival strategy because organisations must manage large data volumes and create targeted representations that align with internal goals. This pragmatic approach was visible in Paper 3, where some forestry firms experienced a sort of "data paralysis" where they were overwhelmed by the sheer volume of data and struggled to decide how to use them or how to engage in digital innovation. Simplification as a result of strategic choices at this level enables organisations to act, even if it narrows the scope of what is represented.

However, when digitalisation moves from the microlevel to societal (macro-) level, the consequences of simplification become more significant, as there is a loss in richness of data. At the microlevel, organisations favour commodity-centred, homogenised data with higher granularity for representing natural environments. Yet at the macrolevel, where equitable and sustainable digitalisation requires data heterogeneity, granularity may be lost to accommodate dominant goals. When the constellation of actors becomes skewed towards those commodification interests, relational and cultural values are often displaced (see Hood & Gabrys, 2024). The exclusion of sustainability or social values, such as data relevant to reindeer husbandry or cultural heritage, thus becomes an outcome of microlevel trade-offs being transferred into broader societal digitalisation efforts. Aaltonen and Tempini (2014) emphasise that the potential of data to inform many unforeseen issues depends on its comprehensiveness and granularity. As such, losing these qualities can undermine the possibility of a digitalisation of natural environments that also foregrounds environmental and social sustainability goals. For societal level digitalisation of natural environments to advance, I propound that granularity and heterogeneity of data should coexist, operating along an axis of detailed data and diverse sources, respectively. In this context, granularity may include detailed data on tree species and forest damages caused by insects, whereas heterogeneity encompasses the breadth of data from environmental groups, cultural perspectives, and recreational users of forests.

Furthermore, my findings reveal that the physicality of the natural environment is not a passive backdrop but plays an active role in shaping how digitalisation unfolds. Forests are spatial, ecological, and concrete units (Gibson & Walford, 2025), which is reflected in my empirical work. In Paper 3 (Projects 2 and 3), for example, improving efficiency in sorting sawmill and pulpwood chips required building a predictive algorithm that used large volumes of data. In the pulp-making process harvested logs are categorised based on their intended end use. The type and quality of the chips determines whether they are best suited for cardboard, printing paper, or packaging paper. Each sawmill has a woodyard that stores these different categories of chips. Having better control over the properties of wood chips therefore gives the pulp industry the opportunity to produce more refined and targeted end-products. The challenge that however emerges as a result of digitalisation is that as the predictive algorithm generates more granular data about the properties of wood chips, the number of classifications multiplies. Research has suggested that big data enables organisations to make informed insights while citing a lack of organisational capabilities as a key challenge to unlocking its potential (Chen et al., 2021). My empirical evidence suggests that material constraints present an additional challenge when data are decoupled from physical objects such as forests. The findings in Paper 3 (Projects 2 and 3) show that as the wood classifications multiply, forestry firms need to store chips in increasingly granular, and separate piles. Yet they lack the warehouse capacity and logistical infrastructure to accommodate the expanding number of categories. Additionally, the forestry firms will incur higher costs from separating wood chips into smaller divisions and improving process control from sorting to delivery. This challenge highlights that the physicality of natural objects imposes limits on digitalisation. Rather than being purely a technical or organisational challenge, the constraints here are material and spatial. Spatiality, as a dimension of physicality (Ghazali, 2006) in this instance, relates to the physical space that objects occupy and the demands this places on how they can be managed and stored. In this way, physicality does not simply enable or constrain what can be captured digitally but actively redirects the scale and direction of digitalisation.

In this thesis, I have discussed how leveraging digital technologies can be seen as a promising asset for digitalising natural environments and contribute to diverse sustainability goals (Merhi et al., 2026; Adebayo et al., 2025). Nonetheless, it would be naïve to get swept up in technofix solutionism (Turnbull et al., 2023) and ignore the dark sides (Schoormann et al., 2025) or negative consequences that digital technologies have on natural environments. At the time of writing this

thesis, we are experiencing the profound impact of digital technologies such as generative artificial intelligence (GenAI) on the climate and environment. Recent reports have shown that the process of training and deploying GenAI models demands a staggering amount of electricity and water[14] [15]. Furthermore, data centres are cropping up to support the boom in GenAI demands with the United Nations Environment Programme (UNEP) highlighting that these data centres are significant contributors to planet-warming greenhouse gas emissions (UNEP, 2025). Thus, the technologies being proposed as tools for addressing environmental challenges may very well be contributing to the problem. This then raises an important question about the paradoxical role that digitalisation plays in relation to sustainability.

On the one hand, digital technologies offer bona fide possibilities to enable more inclusive approaches to nature (Tzachor et al., 2022) and to precisely monitor and manage natural environments which can significantly reduce waste (Adebayo et al., 2025; Zhang & Caretta, 2025). On the other hand, growing scepticism and environmental consequences, including digital technologies contributing to electronic waste (Liu et al., 2023), and the risk of concentrating decision-making power in the hands of dominant actors at the expense of minority actors suggest that digitalisation is not a costless endeavour.

The crux of the issue lies more in how we navigate this paradox than in finding an absolute and singular solution to resolve it. IS research is positioned to play an important role by offering a sociotechnical perspective on the digitalisation of natural environments as a way of thinking intentionally about the choices involved. The sociotechnical perspective draws attention to the social, political, and material dimensions of these processes: who decides what data get collected, whose interests shape how digital representations are constructed, and what consequences follow for different stakeholders whose voices have historically been marginalised. However, this kind of intentionality is not straightforward in practice. As my findings in Paper 4 illustrate, intentionality can be skewed towards the interests of dominant actors which often shape what gets prioritised to be represented in digitalisation efforts. As highlighted in Sections 6.1 and 6.2, in Swedish forestry the decisions about what data to collect and what technologies to deploy can

[14] Going digital- good or bad for the climate? https://climate-pact.europa.eu/articles-and-events/pact-articles/going-digital-good-or-bad-climate-2025-02-19_en

[15] Explained: Generative AI's environmental impact. https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117

reflect and reinforce power relations (see Gabrys et al., 2022). Larger forestry firms with greater resources and capabilities are better positioned to shape the direction of digitalisation, whereas smaller forest owners and environmental actors risk being neglected. The sociotechnical perspective that IS research offers makes these tensions visible rather than obscuring them behind the overly optimistic promise of digital technologies.

Visibility of these tensions is a first step towards more inclusive and intentional approaches to the downsides of digitalisation. However, it would be presumptuous to assume that this will resolve the paradox that exists between digitalisation and sustainability. Digital technologies will continue to play an expanding role in how natural environments are used, monitored, and managed meaning that the tensions between their benefits and costs will inevitably remain. However, approaching these complex tensions as being on a spectrum rather than drawing a hard line between uncritical optimism and complete scepticism is perhaps a way to navigate the digitalisation of natural environments.

7. Conclusions

This thesis has examined how the digitalisation of natural environments materialises across societal, industry, and organisational levels revealing the value conflicts, stakeholder tensions, and sociotechnical interactions that shape this process. Through the conceptual framework developed in this thesis, I have highlighted how different perspectives on data condition what gets abstracted, captured, and represented about the dynamic nature of natural environments. In doing so, I contend that moving from predominantly commodity-oriented approaches of data towards more relational perspectives is essential for creating a more inclusive and encompassing digitalisation of natural environments. In what follows, I discuss the implications of these findings for research and practice. Lastly, I reflect on the limitations of this and the avenues for future research.

7.1 Implications for Research and Practice

This thesis, and the studies appended examine the prominent role of digital technologies in mediating in and addressing grand societal challenges. Based on research investigating the distinct phenomenon of digitalising natural environments through a sensitising device, this thesis provides an analytical lens for studying how digitalisation of such phenomena unfolds across individual, organisational, and societal levels. In doing so, it makes a number of contributions to both IS theory and practice which the following sections elaborate on. The thesis offers four principal theoretical contributions.

First, this thesis contributes to IS research by demonstrating the value of extending scholarly attention beyond artificial objects and organisational settings to the digitalisation of natural environments. IS research has a long tradition of examining how digital technologies reshape the contexts in which they are used (Gradillas & Thomas, 2025; Wessel et al., 2021), yet the predominant focus has been on digital or artificial objects or built environments (e.g., Velsberg et al., 2021; Svahn et al., 2017). Natural environments have been treated as a peripheral setting for using digital technologies rather than as distinct objects of study. This is a significant oversight given the growing urgency of grand societal challenges that the IS field is increasingly poised to address. As previously highlighted in this thesis, climate change and biodiversity loss are among the most pressing global issues of our time, and digital technologies have been formally

recognised by global bodies such as the United Nations as central to how societies can mitigate environmental risks and monitor and manage natural environments (UNDP, 2025). Research disciplines such as ecology and environmental science have been at the helm of the growing movement to explore how digital technologies can be used in natural environments to address societal issues. IS research, with its rich concepts for analysing sociotechnical processes, is equally well positioned to contribute to this effort. Yet, as this thesis has shown, that contribution is still incipient. Although an emerging stream of IS scholarship has begun to turn its attention to the digitalisation of natural environments (e.g., Monteiro & Parmiggiani, 2019; Østerlie et al., 2012; Østerlie & Monteiro, 2020), it remains a largely underexplored area of investigation.

To prevent the digitalisation of natural environment from becoming a chimera in IS research, deliberate scholarly attention is needed. This dissertation takes a step in that direction. By conducting an empirical investigation of digitalisation efforts in the Swedish forestry industry, it demonstrates that IS research has much to offer in understanding how the digitalisation of natural environments unfolds in practice. In doing so, this thesis illustrates how IS scholars can extend their attention beyond artificial objects without abandoning the core concerns of the field. Through my empirical findings, this thesis contributes to IS research by reaffirming its capacity to examine the use of digital technologies from a sociotechnical standpoint and highlight its political and social complexities.

Second, this thesis contributes to research on digitalisation, digital transformation, and digital innovation by enriching these concepts through an empirical investigation focused on natural environments. According to von Krogh et al. (2012), a key means of advancing theoretical concepts is to examine them in novel empirical contexts that challenge existing assumptions and offer alternative frames for viewing phenomena. This thesis shows that natural environments represent a rich phenomenon; unlike the organisational settings where concepts such as digital transformation and innovation have predominantly been explored, natural environments are characterised by distinct objects. These objects introduce challenges that existing conceptualisations of digitalisation and related concepts in IS research have not fully accounted for. By examining digitalisation in the context of Swedish forestry, this thesis shows that perspectives on data – be they commodity, processual, or relational perspectives – have direct implications for the trajectory and outcomes of digitalisation. As my findings have illustrated, data about natural environments that are primarily treated as commodities orient

digitalisation towards economic efficiency and competitive advantage. However, a relational perspective on data foregrounds broader stakeholder interests and extends the scope of what digitalisation encompasses and achieves. This finding adds nuance to recent calls within IS research by Yoo et al. (2024) for greater attention to be paid to data as a driver of digital innovation, by showing that the perspective through which data are understood shape the entire digitalisation process.

Third, the thesis specifically develops and applies a sensitising device to study the digitalisation of natural environments. In attempting to study this phenomenon, it became clear that existing IS frameworks do not adequately account for the distinctive characteristics of natural environments. This echoes the calls of Wessel et al. (2025) for greater attention to the object of digital transformation and related concepts as a means of spurring conceptualisations in the IS research, as well as Jones' (2019) argument that what gets captured and represented into data is a critical question for studying phenomena. Existing research on these dimensions has been fragmented, and this thesis begins to offer a synthesis by drawing on insights from IS research and wider disciplines. The sensitising device comprises of objects of digitalisation, data, and digital representations and was developed to enable a structured analyses of nuances that emerged from my empirical qualitative data. Furthermore, the sensitising device allowed for a more granular understanding of digitalisation in practice. In terms of data, it allowed for unpacking and revealing how within my empirical material the three views (commodity, processual, relational) outlined by Jarvenpaa and Markus (2020) are not mutually exclusive but often interrelated and co-exist within the same digitalisation effort. The sensitising device also shone a light on the relationship between the characteristics of natural objects and digital representations. It made visible how the objects of the digitalisation of the natural environment can drive digitalisation and simultaneously constrain its scale and redirect its trajectory. In this way, I specifically contribute to the IS discourse around digital representations by turning attention to a largely overlooked aspect of digital representations by illustrating through my findings how physicality shapes what can be abstracted, represented, and acted upon digitally.

Fourth, this thesis contributes beyond IS research by offering insights back to the multiple disciples it has drawn on. As previously mentioned, studying the digitalisation of natural environments transcends the boundaries of a single discipline but draws on and intersects with other research disciplines. These fields complemented IS research throughout this thesis, and in turn, the sociotechnical perspective that this thesis

brings offers a comprehensive lens to approach the study of natural environments. In this way, this thesis hopes to contribute to a cross-disciplinary conversation and integrative scholarship on digitalising natural environments, one that IS research is well versed to enrich.

The digitalisation of natural environments is increasingly important to practitioners. This thesis therefore makes three key contributions to practice.

First, the thesis has implications for how managers and organisations approach data in the context of digitalising natural environments. The findings of this thesis reveal that multiple views of data can emerge and co-exist within the same organisation or even the same project. For example, in my empirical studies on the emergence of data spaces in Swedish forestry, the processual and relational perspectives on data were present simultaneously, with actors negotiating how data would be transformed and reused for digitalisation efforts. This suggests that data are rarely viewed from a singular lens in practice as organisations benefit from plurality. Against this background, managers need to be cognisant of and carefully consider which view of data dominates their organisation's digitalisation efforts and whether that view is adequate for the goals they are pursuing. This is not to suggest that the commodity view should be sidelined, as it forms the backbone of commercial operations and remains essential for maintaining competitiveness and profitability. However, this thesis highlights that a purely commodity-centred approach to data is insufficient, particularly given the growing emphasis on sustainability targets in organisational strategies and business models.

Digital sustainability today encompasses not only economic considerations but also environmental and social aspects (Kotlarsky et al., 2023), requiring organisations to meet diverse stakeholder demands and accountability metrics. A pure commodity view on data does not support these broader sustainability ambitions by organisations. In industries such as forestry, which are characterised by long-term planning and where the legitimacy of the industry depends on its ability to balance economic, social, and ecological interests over an extended period of time, a more comprehensive approach to data is necessary. As the empirical findings of this thesis illustrate, the Swedish forestry industry involves multiple actors with competing visions and goals of what the Swedish forest should be used for (Alarcon, 2023), and how data should be collected and used. A relational view of data which considers the diverse values and interests of different stakeholders is therefore key for

upholding the long-term legitimacy of the industry in the setting of natural environments.

Second, this thesis has implications for policy in the digitalisation of natural environments. As Gabrys et al. (2020) highlight, digital technologies are promoting and opening up new forms of participation that change how forest governance that extends beyond experts and traditional institutional actors occurs. In this thesis, digital twins have been proposed as a solution to help bridge conflicting perspectives and create an inclusive space for conversations on natural environments (Blair, 2021; Westerlaken, 2024). However, as the findings of the thesis illustrate, these possibilities are not automatically realised: they depend on the choices about how digital systems are designed, for whom, which data are collected and whose interests are centred in the process. This is particularly important when data on natural environments feeds into policy decisions. Research has shown that data can present an unrepresentative view of natural environments when translated into policy decisions (Gabrys et al., 2022; Pritchard et al., 2022). The findings from this thesis offer a concrete step by proposing design principles for more inclusive digital systems such as digital twins to enable participatory planning of natural environments. These propositions can inform policy and how public funding can be allocated to ensure that research and development efforts for digitalisation are inclusive of a wide array of stakeholders.

Third, although I have drawn research from multiple fields to theoretically frame the phenomenon under study, the digitalisation of natural environments is a process that is ultimately enacted in practice. The thesis argues that to achieve meaningful change, digitalisation requires the active involvement of diverse actors with varying interests and competencies. Rather than being driven by a single group or logic, digitalisation efforts benefit from the convergence of multiple stakeholders who collaboratively explore new ways of leveraging digital solutions. One practical implication is the need for more participatory and inclusive digital systems. As argued in this thesis, digital representations and visualisations should designed in ways that are understandable and accessible to a broad range of actors including industry practitioners, activists, domain experts, with varied backgrounds. Such inclusive systems can support a shared understanding and negotiation on the digitalisation of natural environments. In addition, practitioners can also enact digitalisation through collaborative data practices. Shared data-sharing initiatives can create spaces where researchers and practitioners

can converge and jointly engage with data to inform decision-making and drive digitalisation in natural environments.

7.2 Limitations and Future Research

As with all other research, this thesis is not without limitations. In this section, I discuss the limitations of my research and outline how they open up avenues for future research.

In this thesis, studying a distinct phenomenon such as digitalising natural environments through a sensitising device offered many contributions to research as highlighted in Section 7.1, but it also has its limitations. Although the sensitising device has been analytically valuable in providing a structured lens for studying this phenomenon, it was developed through, and grounded in, a single empirical setting. The actors and stakeholder interests examined in this thesis are, to a certain extent, grounded in the specifics of the Swedish forestry context. This limitation is further compounded by the nature of interpretive research. As Walsham (2006) notes, theory development in interpretive research is shaped by the inherently subjective choices that researchers make on what gets either foregrounded or understated. The three core elements of the sensitising device – objects of digitalisation, data, and digital representations – were developed inductively to capture and understand the empirical nuances present in the qualitative material analysed in the appended papers. These elements therefore reflect an analytical focus that emerged from within the context of Swedish forestry, meaning the sensitising device is by no means exhaustive. As such, certain aspects of digitalising natural environments may fall outside the scope afforded by the sensitising device and escape analytical attention. Following Yin's (2014) distinction between analytical and statistical generalisation, the findings of this thesis are not intended to support statistical inference to a broader population sample. Instead, the contribution lies in analytical generalisation, whereby the empirical insights from this thesis can be used to inform and develop theory about the sociotechnical interactions involved in the digitalisation of natural environments.

To strengthen the validity and further develop theoretical contributions, future research can apply, test, and refine the sensitising device in new empirical settings. In line with Lee and Baskerville's (2003) argument that theory should be confirmed and extended by applying it to different contexts, researchers should not simply adopt the sensitising device as it stands. Instead, it should be adapted, challenged, and expanded as it is

used in other domains such as marine environments and agriculture. Such cumulative work would identify facets of digitalisation that were not visible in this thesis and contribute to the small yet emerging development of theory in IS research on the digitalisation of natural environments.

This thesis has adopted an interpretive stance and qualitative methods to approach the appended papers, which introduces methodological limitations. Interpretive approaches are well suited for developing rich understandings of complex and emerging phenomena (Klein & Myers, 1999) such as the digitalisation of natural environments. Accordingly, this thesis contributes what Walsham (1995) describes as "rich insight" into how actors enact digitalisation in the specific empirical context of Swedish forestry. The interpretive stance also foregrounds subjective meanings, and this thesis acknowledges that the digitalisation process is perceived differently by different actors (Klein & Myers, 1999). Moreover, Monteiro et al. (2022) emphasise that qualitative researchers become part of the research setting and inevitably bring prior assumptions or perspectives that shape analysis during the research process. Throughout this thesis, I have endeavoured to maintain reflexivity and have shown how my embeddedness in Swedish forestry context shaped my analytical focus. Nevertheless, the findings from my empirical work reflect situated understandings rather than exhaustive representations of the phenomenon (Lee, 1991). Furthermore, while qualitative methods help develop deep understanding of a phenomenon and offer rich insights (Walsham, 1995), it has limitations in terms of the scope of what can be examined. As such, these limitations open up opportunities for further research. Quantitative and mixed methods could strengthen the generalisability of the theoretical contributions this thesis makes. A mixed method approach will specifically allow for exploring and confirming research inquiry that helps offset the weaknesses of a single research method and provide complementary strengths that individual methods may not offer (Venkatesh et al., 2013). This thesis therefore encourages IS researchers to explore the digitalisation of natural environments using a broad range of methodological approaches.

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